

Q1: A mosquito is moving with a velocity $\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$ m/s and accelerating in uniform conditions. What will be the direction of mosquito after 2 s?

- (A) $\tan^{-1} \left(\frac{2}{3} \right)$ from y-axis
- (B) $\tan^{-1} \left(\frac{5}{2} \right)$ from x axis
- (C) $\tan^{-1} \left(\frac{2}{3} \right)$ from x axis
- (D) $\tan^{-1} \left(\frac{5}{2} \right)$ from y axis

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(C) $\tan^{-1} \left(\frac{2}{3} \right)$ from x axis

(D) $\tan^{-1} \left(\frac{5}{2} \right)$ from y axis

Solution:

Given:

$$\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$$

$$\vec{v}_{\text{at } t=2} = 2\hat{i} + 6\hat{j} + 9\hat{k}$$

∴ Angle made by direction of motion of mosquito will be,

$$\cos^{-1} \frac{2}{11} (\text{from x - axis}) = \tan^{-1} \frac{\sqrt{117}}{2}$$

$$\cos^{-1} \frac{6}{11} (\text{from y - axis}) = \tan^{-1} \frac{\sqrt{85}}{6}$$

$$\cos^{-1} \frac{9}{11} (\text{from z - axis}) = \tan^{-1} \frac{\sqrt{40}}{9}$$

None of the option is matching.

Hence this question should be bonus.

Q2: A charge Q is moving $\vec{dl}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$.

- (A) -1
- (B) Zero
- (C) Infinite
- (D) 1

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Solution:

Since force on a point charge by magnetic field is always perpendicular to $\vec{v}$ $[\vec{F} = q\vec{V} \times \vec{B}]$

$\therefore$ Work by magnetic force on the point charge is zero

Q3: The half-life of Au^{198} is 2.7 days. The activity of 1.50 mg of Au^{198} if its atomic weight is 198 g mol^{-1} is, ($N_A = 6 \times 10^{23}/\text{mol}$)

- (A) 252 Ci
- (B) 535 Ci
- (C) 240 Ci
- (D) 357 Ci

Q3: The half-life of Au^{198} is 2.7 days. The activity of 1.50 mg of Au^{198} if its atomic weight is 198 g mol⁻¹ is, ($N_A = 6 \times 10^{23}$ /mol)

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(B) 535 Ci

(C) 240 Ci

(D) 357 Ci

Solution:

$$A = \lambda N$$

$$N = n N_A \left(t_{1/2} = \frac{\ln 2}{\lambda} \right)$$

$$N = \left(\frac{1.5 \times 10^{-3}}{198} \right) N_A$$

$$A = \left(\frac{\ln 2}{t_{1/2}} \right) N$$

$$1 \text{ curie} = 3.7 \times 10^{10} \text{ Bq}$$

$$A = 365 \text{ Bq}$$

Q4: In order to determine the Young's Modulus of a wire of radius 0.2 cm (measured using a scale of least count = 0.001 cm) and length 1 m (measured using a scale of least count = 1 mm), a weight of mass 1 kg (measured using a scale of least count = 1 g) was hanged to get the elongation of 0.5 cm (measured using a scale of least count 0.001 cm). What will be the percentage error in the value of Young's Modulus determined by this experiment ?

- (A) 0.14 %
- (B) 1.4%
- (C) 9%
- (D) 0.9%

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Solution:

$$\begin{aligned}
 Y &= \frac{\text{Stress}}{\text{Strain}} = \frac{FL}{\Delta l} = \frac{mg.L}{\pi R^2 \ell} \\
 \frac{\Delta Y}{Y} &= \frac{\Delta m}{m} + \frac{\Delta L}{L} + 2 \cdot \frac{\Delta R}{R} + \frac{\Delta \ell}{\ell} \\
 \frac{\Delta Y}{Y} \times 100 &= 100 \left[\frac{1}{1000} + \frac{1}{1000} + 2 \left(\frac{0.001}{0.2} \right) + \frac{0.001}{0.5} \right] \\
 &= \frac{1}{10} + \frac{1}{10} + 1 + \frac{1}{5} = \frac{14}{10} = 1.4\%
 \end{aligned}$$

Q5: A resistor develops 500 J of thermal energy in 20 s when a current of 1.5 A is passed through it. If the current is increased from 1.5 A to 3 A, what will be the energy developed in 20 s.

- (A) 2000 J
- (B) 500 J
- (C) 1000 J
- (D) 1500 J

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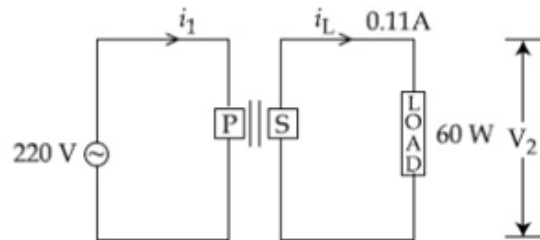
Solution:

$$500 = (1.5)^2 \times R \times 20$$

$$E = (3)^2 \times R \times 20$$

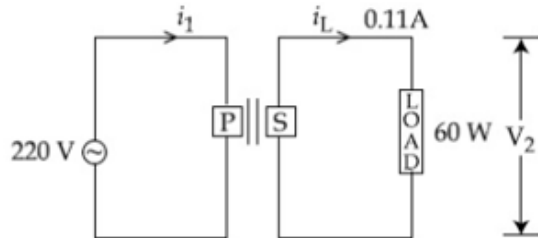
$$E = 2000J$$

Q6: For the given circuit, comment on the type of transformer used.



- (A) Step - up transformer
- (B) Step down transformer
- (C) Auto transformer
- (D) Auxiliary transformer

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Solution:

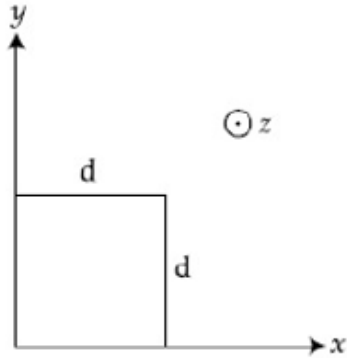
$$V_S = \frac{P}{i} = \frac{60}{0.11} = 545.45$$

$$V_P = 220$$

$$V_S > V_P$$

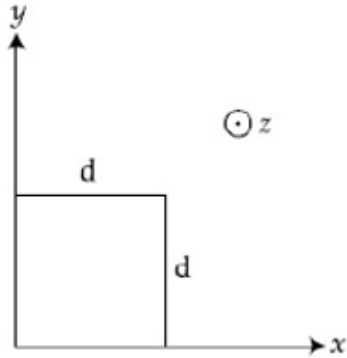
⇒ Step up transformer

Q7: The magnetic field in a region is given by $\vec{B} = B_0 \left(\frac{x}{a} \right) \hat{k}$. A square loop of side d is placed with its edges along the x and y axes. The loop is moved with a constant velocity $\vec{v} = v_0 \hat{i}$. The emf induced in the loop is :



- (A) $\frac{B_0 v_0^2 d}{2a}$
- (B) $\frac{B_0 v_0 d^2}{a}$
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- (D) $\frac{B_0 v_0 d^2}{2a}$

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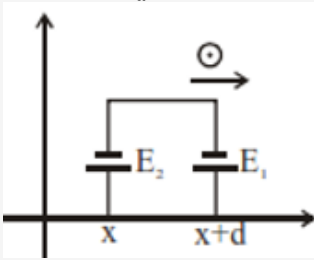
Solution:

$$E_1 = \frac{B_0(x+d)}{a} v_0 d$$

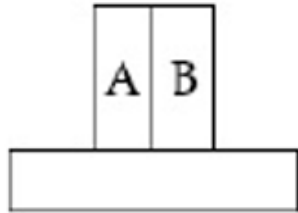
$$E_2 = \frac{B_0(x)}{a} v_0 d$$

$$E_{net} = E_1 - E_2$$

$$E_{net} = \frac{B_0 v_0 d^2}{a}$$

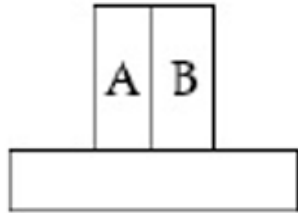


Q8: A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that of metal B. When the bimetallic strip is placed in a cold bath, it will :



- (A) Not bend but shrink
- (B) Bend towards the left
- (C) Bend towards the right
- (D) Neither bend nor shrink

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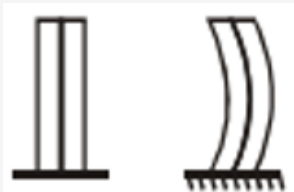
- (A) Not bend but shrink
- (B) Bend towards the left
- (C) Bend towards the right
- (D) Neither bend nor shrink

Solution:

$$\alpha_A > \alpha_B$$

Length of both strips will decrease

$$\Delta L_A > \Delta L_B$$



Q9: Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure $1.01 \times 10^5 \text{ Pa}$. Assume the molecular diameter 0.3 nm and the gas is ideal. ($k = 1.38 \times 10^{-23} \text{ JK}^{-1}$)

- (A) 86 nm
- (B) 102 nm
- (C) 32 nm
- (D) 58 nm

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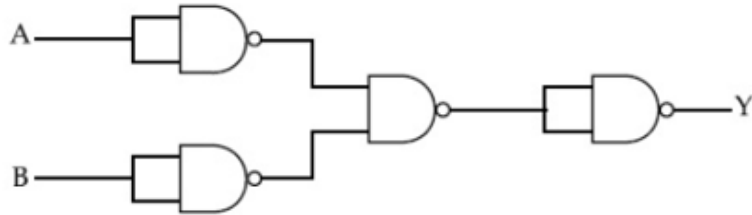
Solution:

$$\lambda = \frac{kT}{\sqrt{2}\pi d^2 P}$$

putting all the values and solving them

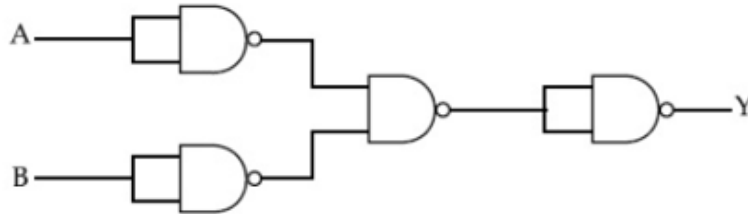
$$\lambda = 102 \text{ nm}$$

Q10: The following logic gate is equivalent to:



- (A) AND Gate
- (B) NAND Gate
- (C) NOR Gate
- (D) OR Gate

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- (A) AND Gate
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- (C) NOR Gate
- (D) OR Gate

Solution:

Truth table for the given logic gate:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The truth table is similar to that of a NOR gate.

Q11: Red light differs from blue light as they have :

- (A) Same frequencies and same wavelengths
- (B) Different frequencies and same wavelengths
- (C) Same frequencies and different wavelengths
- (D) Different frequencies and different wavelengths

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Solution:

Red light and blue light have different wavelength and different frequency.

Q12: Statement 1: A cyclist is moving on an unbanked road with a speed of 7 kmh^{-1} and takes a sharp circular turn along a path of radius of 2m without reducing the speed. The static friction coefficient is 0.2. The cyclist will not slip and pass the curve. ($g = 9.8 \text{ m/s}^2$).

Statement II: If the road is banked at an angle of 45° , cyclist can cross the curve of 2m radius with the speed of 18.5 kmh^{-1} without slipping. In the light of the above statements, choose the correct answer from the options given below.

- (A) Both statement I and statement II are false
- (B) Both statement I and statement II are true
- (C) Statement I is incorrect and statement II is correct
- (D) Statement I is correct and statement II is incorrect

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Solution:

Statement I:

$$V_{\max} = \sqrt{\mu Rg} = \sqrt{(0.2) \times 2 \times 9.8}$$

$$V_{\max} = 1.97m/s$$

$$7km/h = 1.944m/s$$

Speed is lower than $V_{\max}$, hence it can take safe turn.

Statement I:

$$V_{\max} = \sqrt{Rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]}$$

$$= \sqrt{2 \times 9.8 \left[\frac{1+0.2}{1-0.2} \right]} = 5.42m/s$$

$$18.5km/h = 5.14m/s$$

Speed is lower than $V_{\max}$, hence it can take safe turn.

Q13: Amplitude of a mass-spring system, which is executing simple harmonic motion decreases with time. If mass = 500g. Decay constant = 20 g/s then how much time is required for the amplitude of the system to drop to half of its initial value? ($\ln 2 = 0.693$)

- (A) 17.32
- (B) 34.65
- (C) 0.034
- (D) 15.01

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Solution:

$$\begin{aligned} A &= A_0 e^{-\gamma t} = A_0 e^{-\frac{bt}{2m}} \\ \frac{A_0}{2} &= A_0 e^{-\frac{bt}{2m}} \\ \frac{bt}{2m} &= \ln 2 \\ t &= \frac{2m}{b} \ln 2 = \frac{2 \times 500 \times 0.693}{20} \\ t &= 34.65 \text{ second} \end{aligned}$$

Q14: The de-Broglie wavelength associated with an electron and a proton were calculated by accelerating them through same potential of 100 V. What should nearly be the ratio of their wavelengths ? ($m_p = 1.00727u$; $m_e = 0.00055u$)

- (A) 43 : 1
- (B) $(1860)^2$: 1
- (C) 41.4 : 1
- (D) 1860 : 1

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- (A) 43 : 1
- (B) $(1860)^2 : 1$
- (C) 41.4 : 1
- (D) 1860 : 1

Solution:

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$
$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1}}$$
$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}} = \sqrt{1831.4} = 42.79$$

Q15: Find out the surface charge density at the intersection of point $x = 3$ m plane and x-axis, in the region of uniform line charge of 8 nC/m lying along the z-axis in free space.

- (A) 0.07 nC m⁻²
- (B) 47.88 C/m
- (C) 0.424 nC m⁻²
- (D) 4.0 nC m⁻²

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- (C) 0.424 nC m^{-2}
- (D) 4.0 nC m^{-2}

Solution:

$$\frac{2K\lambda}{r} = \frac{\sigma}{\epsilon_0} \quad (x = 3\text{m})$$
$$\sigma = 0.424 \times 10^{-9} \frac{\text{C}}{\text{m}^2}$$

Q16: Two identical antennas mounted on identical towers are separated from each other by a distance of 45 km. What should nearly be the minimum height of receiving antenna to receive the signals in line of sight ?

(Assume radius of earth is 6400 km)

- (A) 39.55 m
- (B) 158.2 m
- (C) 79.1 m
- (D) 19.77 m

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Solution:

$$D = 2\sqrt{2Rh}$$
$$h = \frac{D^2}{8R} = \frac{45^2}{8 \times 6400} km \cong 39.55m$$

Q17: What will be the nature of flow of water from a circular tap, when its flow rate increased from 0.18 L/min to 0.48 L/min ? The radius of the tap and viscosity of water are 0.5 cm and 10^{-3} Pa s. respectively.

(Density of water : 10^3 kg/m³)

- (A) Remains turbulent flow
- (B) Steady flow to unsteady flow
- (C) Unsteady to steady flow
- (D) Remains steady flow

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Solution:

The nature of flow is determined by Reynolds Number.

$$R_e = \frac{\rho v D}{\eta}$$

$$\left[\begin{array}{l} \rho \rightarrow \text{density of fluid; } \eta \rightarrow \text{coefficient of viscosity} \\ v \rightarrow \text{velocity of flow} \\ D \rightarrow \text{Diameter of pipe} \end{array} \right]$$

From NCERT

If $R_e < 1000 \rightarrow$ flow is steady

$1000 < R_e < 2000 \rightarrow$ flow becomes unsteady

$R_e > 2000 \rightarrow$ flow is turbulent

$$R_{e \text{ initial}} = 10^3 \times \frac{0.18 \times 10^{-3}}{\pi \times (0.5 \times 10^{-2})^2 \times 60} \times \frac{1 \times 10^{-2}}{10^{-3}}$$

$$= 382.16$$

$$R_{e \text{ final}} = 10^3 \times \frac{0.48 \times 10^{-3}}{\pi \times (0.5 \times 10^{-2})^2 \times 60} \times \frac{1 \times 10^{-2}}{10^{-3}}$$

$$= 1019.09$$

Q18: The refractive index of a converging lens is 1.4. What will be the focal length of this lens if it is placed in a medium of same refractive index ?(Assume the radii of curvature of the faces of lens are R_1 and R_2 respectively).

(A) $\frac{R_1 R_2}{R_1 - R_2}$

(B) zero

(C) Infinite

(D) 1

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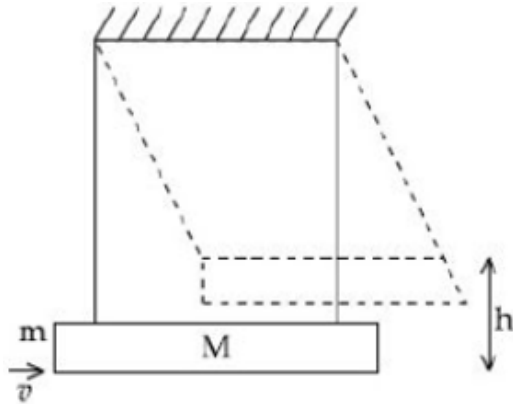
(D) 1

Solution:

$$\frac{1}{F} = \left[\frac{\mu_L}{\mu_S} - 1 \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

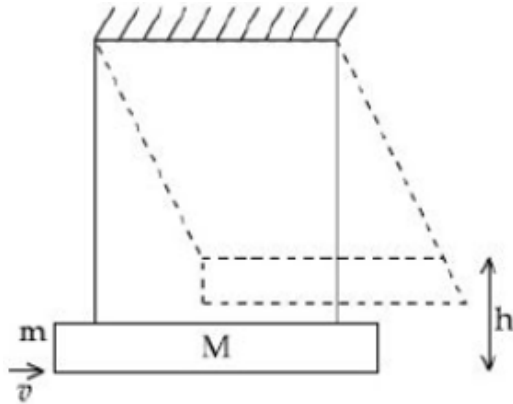
$$\text{If } \mu_L = \mu_S \Rightarrow \frac{1}{F} = 0 \Rightarrow F = \infty$$

Q19: A large block of wood of mass $M = 5.99 \text{ kg}$ is hanging from two long massless cords. A bullet of mass $m = 10 \text{ g}$ is fired into the block and gets embedded in it. The (block + bullet) then swing upwards, their centre of mass rising a vertical distance $h = 9.8 \text{ cm}$ before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is : (take $g = 9.8 \text{ ms}^{-2}$)



- (A) 821.4 m/s
- (B) 841.4 m/s
- (C) 831.4 m/s
- (D) 811.4 m/s

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Solution:

From energy conservation,

[after bullet gets embeded till the system comes momentarily at rest]

$$(M + m) g h = \frac{1}{2} (M + m) v_1^2 \text{ [} v_1 \text{ is velocity after collision]}$$

$$\therefore v_1 = \sqrt{2gh}$$

Applying momentum conservation, (just before and just after collision)

$$v = \left(\frac{M+m}{m} \right) v_1 = \frac{6}{10 \times 10^{-3}} \times \sqrt{2 \times 9.8 \times 9.8 \times 10^{-2}}$$
$$\approx 831.55 \text{ m/s}$$

Q20: Calculate the time interval between 33% decay and 67% decay if half-life of a substance is 20 minutes.

- (A) 20 minutes
- (B) 60 minutes
- (C) 13 minutes
- (D) 40 minutes

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Solution:

$$N_1 = N_0 e^{-\lambda t_1}$$

$$\frac{N_1}{N_0} = e^{-\lambda t_1}$$

$$0.67 = e^{-\lambda t_1}$$

$$\ln(0.67) = -\lambda t_1$$

$$N_2 = N_0 e^{-\lambda t_2}$$

$$\frac{N_2}{N_0} = e^{-\lambda t_2}$$

$$0.33 = e^{-\lambda t_2}$$

$$\ln(0.67) - \ln(0.33) = \lambda t_1 - \lambda t_2$$

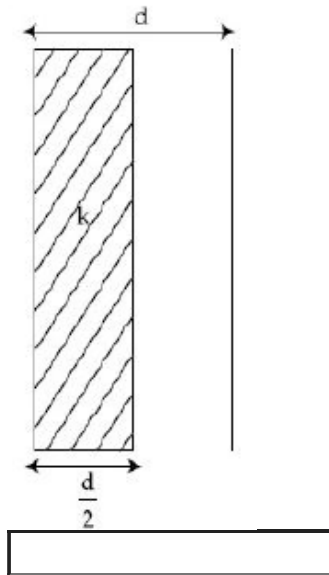
$$\lambda(t_1 - t_2) = \ln\left(\frac{0.67}{0.33}\right)$$

$$\lambda(t_1 - t_2) \cong \ln 2$$

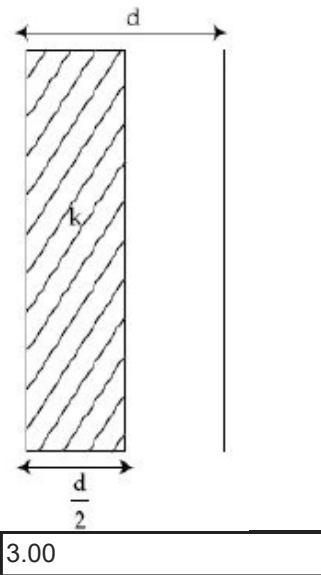
$$t_1 - t_2 \simeq \frac{\ln 2}{\lambda} = t_{1/2}$$

$$\text{Half life} = t_{1/2} = 20 \text{ minutes}$$

Q21: In a parallel plate capacitor set up, the plate area of capacitor is 2 m^2 and the plates are separated by 1 m . If the space between the plates are filled with a dielectric material of thickness 0.5 m and area 2 m^2 (see fig) the capacitance of the set-up will be _____ ϵ_0 . (Dielectric constant of the material = 3.2) (Round off to the Nearest Integer)



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3.00

Solution:

$$\begin{aligned}
 C &= \frac{\epsilon_0 A}{\frac{d}{2k} + \frac{d}{2}} = \frac{2\epsilon_0 A}{\frac{d}{k} + d} \\
 &= \frac{2 \times 2\epsilon_0}{\frac{1}{3.2} + 1} = \frac{4 \times 3.2}{4.2} \epsilon_0 \\
 &= 3.04\epsilon_0
 \end{aligned}$$

Q22: If one wants to remove all the mass of the earth to infinity in order to break it up completely. The amount of energy that needs to be supplied will be $\frac{x}{5} \frac{GM^2}{R}$ where x is _____
(Round off to the Nearest Integer)
(M is the mass of earth; R is the radius of earth; G is the gravitational constant)

Q22: If one wants to remove all the mass of the earth to infinity in order to break it up completely. The amount of energy that needs to be supplied will be $\frac{x}{5} \frac{GM^2}{R}$ where x is _____
(Round off to the Nearest Integer)
(M is the mass of earth; R is the radius of earth; G is the gravitational constant)

3.00

Solution:

$$\begin{aligned}\text{Energy given} &= U_f - U_i \\ &= 0 - \left(-\frac{3}{5} \frac{GM^2}{R} \right) \\ &= \frac{3}{5} \frac{GM^2}{R} \\ x &= 3\end{aligned}$$

Q23: For an ideal heat engine, the temperature of the source is 127°C . In order to have 60% efficiency the temperature of the sink should be _____ $^{\circ}\text{C}$. (Round off to the Nearest Integer)

Q23: For an ideal heat engine, the temperature of the source is 127°C . In order to have 60% efficiency the temperature of the sink should be _____ $^{\circ}\text{C}$. (Round off to the Nearest Integer)

-113.00

Solution:

$$n = 0.60 = 1 - \frac{T_L}{T_H}$$

$$\begin{aligned}\frac{T_L}{T_H} = 0.4 \Rightarrow T_L &= 0.4 \times 400 \\ &= 160\text{K} \\ &= -113^{\circ}\text{C}\end{aligned}$$

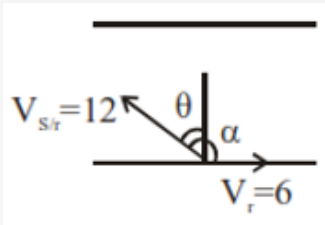
Q24: A swimmer can swim with velocity of 12 km/h in still water. Water flowing in a river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____°C. (Round off to the Nearest Integer) (Find the angle in degrees)

Q24: A swimmer can swim with velocity of 12 km/h in still water. Water flowing in a river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____°C. (Round off to the Nearest Integer) (Find the angle in degrees)

120.00

Solution:

$$12 \sin \theta = v_r$$



$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\therefore \alpha = 120^\circ$$

Q25: A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass as 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be _____ $^\circ$ (in degree). (Round off to the Nearest Integer)

Q25: A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass as 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be _____ $^\circ$ (in degree). (Round off to the Nearest Integer)

12.00

Solution:

$$\omega_1 = 0.02; \mu_1 = 1.5; \omega_2 = 0.03; \mu_2 = 1.6$$

Achromatic combination

$$\therefore \theta_{\text{net}} = 0$$

$$\theta_1 - \theta_2 = 0$$

$$\theta_1 = \theta_2$$

$$\omega_1 \delta_1 = \omega_2 \delta_2$$

$$\&\delta_{\text{net}} = \delta_1 - \delta_2 = 2^\circ$$

$$\delta_1 \left(1 - \frac{\omega_1}{\omega_2}\right) = 2^\circ$$

$$\delta_1 \left(1 - \frac{2}{3}\right) = 2^\circ$$

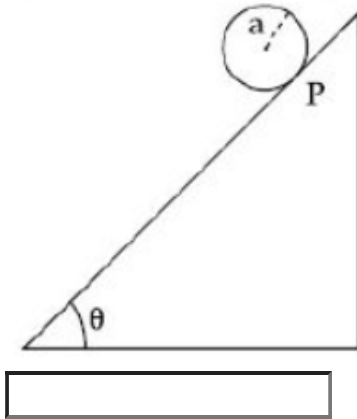
$$\delta_1 = 6^\circ$$

$$\delta_1 = (\mu_1 - 1) A_1$$

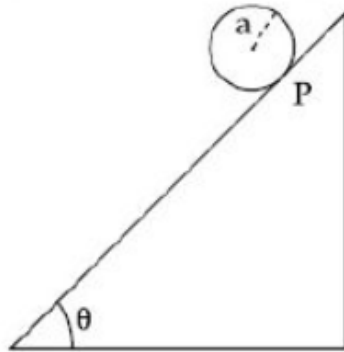
$$6^\circ = (1.5 - 1) A_1$$

$$A_1 = 12^\circ$$

Q26: A solid disc of radius 'a' and mass 'm' rolls down without slipping on an inclined plane making an angle θ with the horizontal. The acceleration of the disc will be $\frac{2}{b} g \sin \theta$ where b is _____. (Round off to the Nearest Integer) (g = acceleration due to gravity; θ = angle as shown in figure)



Q26: A solid disc of radius 'a' and mass 'm' rolls down without slipping on an inclined plane making an angle θ with the horizontal. The acceleration of the disc will be $\frac{2}{b}g \sin \theta$ where b is _____. (Round off to the Nearest Integer) (g = acceleration due to gravity; θ = angle as shown in figure)



3.00

Solution:

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2g}{3} \sin \theta$$

b = 3

Q27: The energy dissipated by a resistor is 10 mJ in 1 s when an electric current of 2 mA flows through it. The resistance is _____ Ω . (Round off to the Nearest Integer)

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Solution:

$$Q = i^2 R T$$
$$R = \frac{Q}{i^2 t} = \frac{10 \times 10^{-3}}{4 \times 10^{-6} \times 1} = 2500 \Omega$$

Q28: A force $\vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ is applied on an intersection point of $x = 2$ plane and x -axis. The magnitude of torque of this force about a point $(2, 3, 4)$ is _____. (Round off to the Nearest Integer)

Q28: A force $\vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ is applied on an intersection point of $x = 2$ plane and x -axis. The magnitude of torque of this force about a point $(2, 3, 4)$ is _____. (Round off to the Nearest Integer)

20.00

Solution:

$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} \\ \vec{r} &= (2\hat{i}) - (2\hat{i} + 3\hat{j} + 4\hat{k}) = -3\hat{j} - 4\hat{k} \\ \vec{F} &= 4\hat{i} + 3\hat{j} + 4\hat{k} \\ \vec{\tau} &= \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -4 \\ 4 & 3 & 4 \end{vmatrix} \\ &= \hat{i}(-12 + 12) - \hat{j}(0 + 16) + \hat{k}(0 + 12) \\ &= -16\hat{j} + 12\hat{k} \\ \therefore |\vec{\tau}| &= \sqrt{16^2 + 12^2} = 20\end{aligned}$$

Q29: A body of mass 2 kg moves under a force of $(2\hat{i} + 3\hat{j} + 5\hat{k})$ N. It starts from rest and was at the origin initially. After 4 s, its new coordinates are (8, b, 20). The value of b is _____.
(Round off to the Nearest Integer)

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Solution:

$$\begin{aligned}\vec{a} &= \frac{\vec{F}}{m} = \frac{2\hat{i}+3\hat{j}+5\hat{k}}{2} \\ &= \hat{i} + 1.5\hat{j} + 2.5\hat{k} \\ \vec{r} &= \vec{u}t + \frac{1}{2}\vec{a}t^2 \\ &= 0 + \frac{1}{2}(\hat{i} + 1.5\hat{j} + 2.5\hat{k})(16) \\ &= 8\hat{i} + 12\hat{j} + 20\hat{k} \\ b &= 12\end{aligned}$$

Q30: A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is

$\frac{x}{3} L \sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

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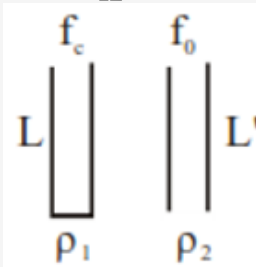
$\frac{x}{3} L \sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

4.00

Solution:

$$f_c = f_o$$

$$\frac{3V_C}{4L} = \frac{2V_0}{2L'}$$



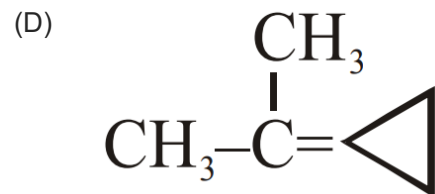
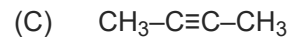
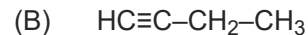
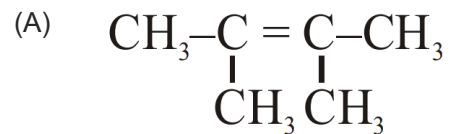
$$\frac{3V_C}{4L} = \frac{V_0}{L'}$$

$$L' = \frac{4L}{3} \frac{V_0}{V_C} = \frac{4L}{3} \sqrt{\frac{B \cdot \rho_1}{\rho_2 \cdot B}} \text{ (B is bulk modulus)}$$

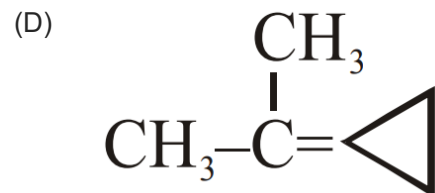
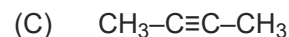
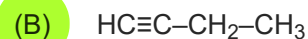
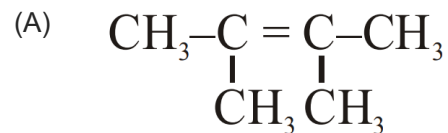
$$= \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

$$x = 4$$

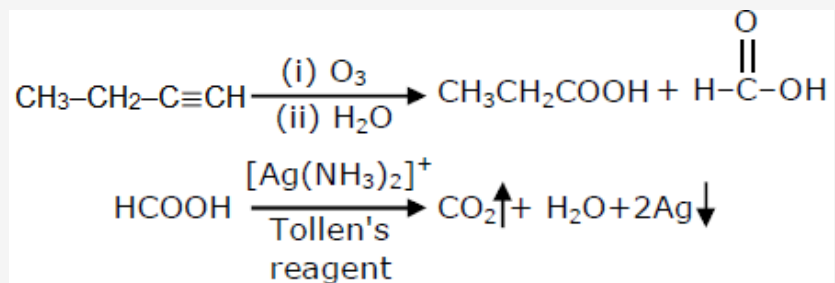
Q31: An unsaturated hydrocarbon X on ozonolysis gives A. Compound A when warmed with ammonical silver nitrate forms a bright silver mirror along the sides of the test tube. The unsaturated hydrocarbon X is:



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Solution:



Q32: Match List-I with List-II:

List-I	List-II
Test/Reagents/Observation(s)	Species detected
(a) Lassaigne's Test	(i) Carbon
(b) Cu(II) oxide	(ii) Sulphur
(c) Silver nitrate	(iii) N, S, P and halogen
(d) The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv) Halogen Specifically

Tile correct match is:

- (A) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (B) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (C) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (D) (a)-(i), (b)-(ii), (c)-(iv), (d)- (iii)

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Tile correct match is:

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- (B) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (C) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (D) (a)-(i), (b)-(ii), (c)-(iv), (d)- (iii)

Solution:

(a) Lassaigne's Test	(iii) N, S, P and Halogen
(b) Cu(II) Oxide	(i) Carbon
(c) AgNO ₃	(iv) Halogen specifically
(d) Sodium fusion extract given black precipitate with acetic acid and lead acetate (CH ₃ COOH/(CH ₃ COO) ₂ Pb)	(ii) Sulphur

Q33: **Statement I:** Sodium hydride can be used as an oxidising agent.
Statement II: The lone pair of electrons on nitrogen in pyridine makes it basic.
Choose the correct answer from the options given below:

- (A) Statement I is true but statement II is false
- (B) Both statement I and statement II are false
- (C) Statement I is false but statement II is true
- (D) Both statement I and statement II are true

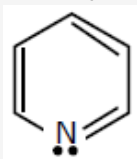
Q33: Statement I: Sodium hydride can be used as an oxidising agent.
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Choose the correct answer from the options given below:

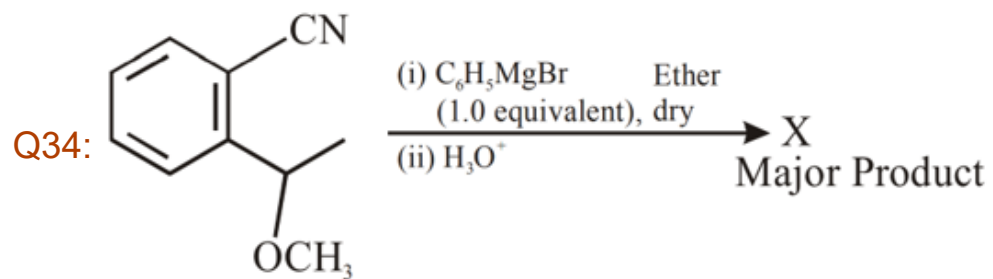
- (A) Statement I is true but statement II is false
- (B) Both statement I and statement II are false
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Solution:

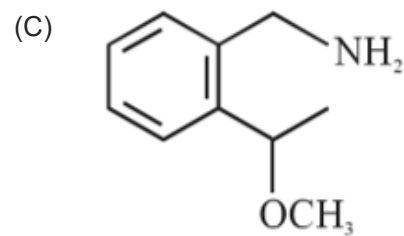
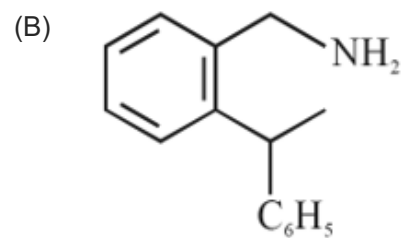
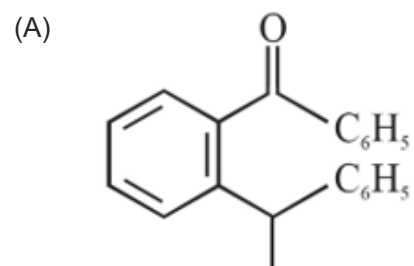
⇒ NaH is used as reducing agent.

⇒ The lp on nitrogen in pyridine makes it basic

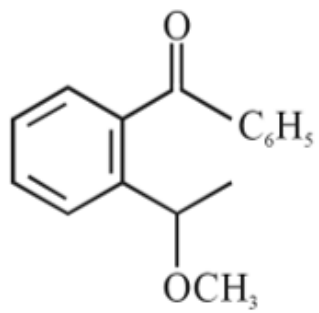


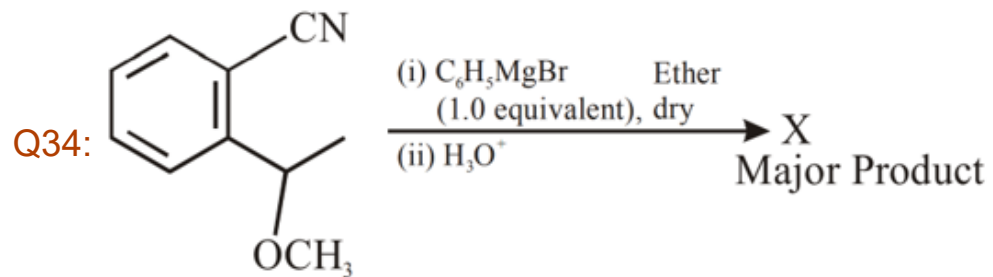


The structure of X is:



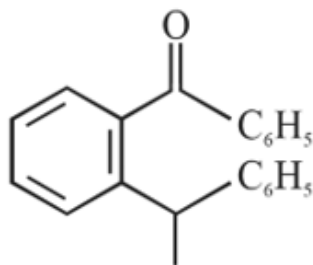
(D)



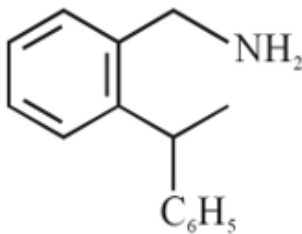


The structure of X is:

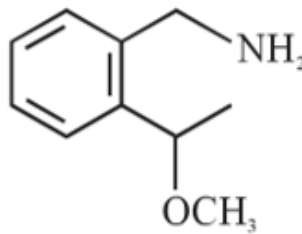
(A)



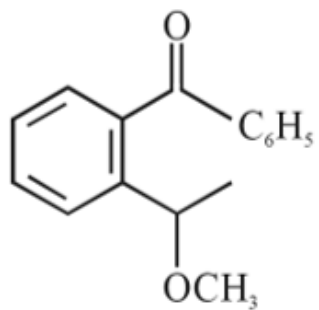
(B)



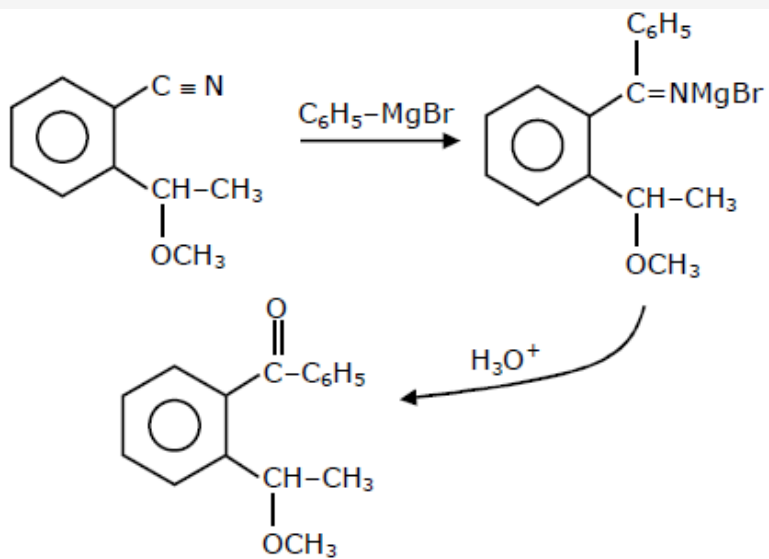
(C)



(D)



Solution:



Q35: The characteristics of elements X, Y and Z with atomic numbers, respectively, 33, 53 and 83 are:

- (A) X and Y are metalloids and Z is a metal
- (B) X, Y and Z are metals
- (C) X and Z are non-metals and Y is a metalloid
- (D) X is a metalloid, Y is a non-metal and Z is a metal

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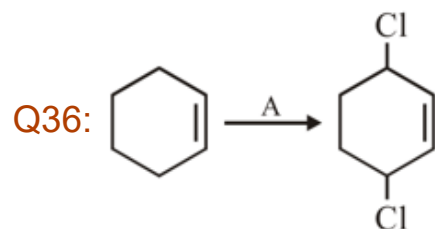
Solution:

Atomic No., Element

(1) 33 → As (Metalloid)

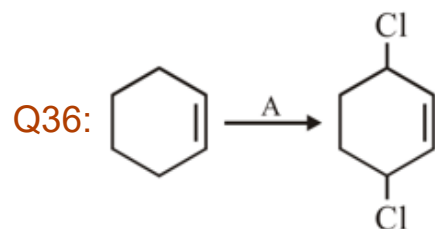
(2) 53 → I (Non metal)

(3) 83 → Bi (Metal)



Identify the reagent(s) 'A' and condition(s) for the reaction

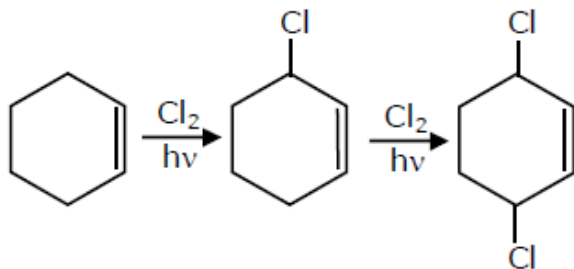
- (A) A = HCl ; Anhydrous AlCl_3
- (B) A = HCl, ZnCl_2
- (C) A = Cl_2 ; dark, Anhydrous AlCl_3
- (D) A = Cl_2 ; UV light



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Solution:



Q37: The INCORRECT statement regarding the structure of C_{60} is :

- (A) The six-membered rings are fused to both six and five-membered rings.
- (B) The five-membered rings are fused only to six-membered rings.
- (C) It contains 12 six-membered rings and 24 five-membered rings.
- (D) Each carbon atom forms three sigma bonds.

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- (D) Each carbon atom forms three sigma bonds.

Solution:

it contains 12 five membered ring & 20 six membered ring

Q38: Identify the elements X and Y using the ionisation energy values given below :

Ionization energy (kJ/mol)

	1st	2nd
X	495	4563
Y	731	1450

- (A) X = F; Y = Mg
- (B) X = Mg; Y = F
- (C) X = Mg; Y = Na
- (D) X = Na; Y = Mg

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- (D) X = Na; Y = Mg

Solution:

2nd I. E of Alkali metals is higher than their respective period.

Q39: The secondary structure of protein is stabilised by

- (A) glycosidic bond
- (B) Hydrogen bonding
- (C) Peptide bond
- (D) Van der Waals forces

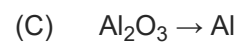
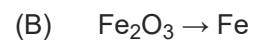
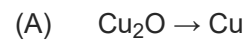
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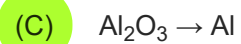
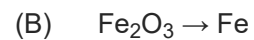
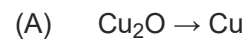
Solution:

The secondary structure of protein stabilised by H-bonding.

Q40: Which of the following reduction reaction CANNOT be carried out with coke?



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Solution:

Al is extracted by electrolytic reduction of Al_2O_3

Q41: The exact volumes of 1 M NaOH solution required to neutralise 50 mL of 1 M H_3PO_3 solution and 100 mL of 2 M H_3PO_2 solution, respectively, are:

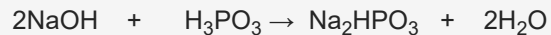
- (A) 100 mL and 50 mL
- (B) 50 mL and 50 mL
- (C) 100 mL and 200 mL
- (D) 100 mL and 100 mL

Q41: The exact volumes of 1 M NaOH solution required to neutralise 50 mL of 1 M H_3PO_3 solution and 100 mL of 2 M H_3PO_2 solution, respectively, are:

- (A) 100 mL and 50 mL
- (B) 50 mL and 50 mL
- (C) 100 mL and 200 mL
- (D) 100 mL and 100 mL

Solution:

(1)



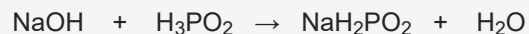
100m mole 50m mole

$$100\text{m mole} = M \times V_{\text{ml}}$$

$$100\text{m mole} = 1 \times V_{\text{ml}}$$

$$V_{\text{ml}} = 100 \text{ ml}$$

(2)



200 m mole 200 m mole

$$200\text{m mole} = M \times V_{\text{ml}}$$

$$V_{\text{ml}} = 200 \text{ ml}$$

Q42: Which of the following polymer is used in the manufacture of wood laminates?

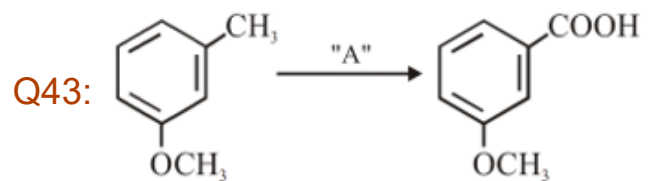
- (A) Melamine formaldehyde resin
- (B) cis-poly isoprene
- (C) Phenol and formaldehyde resin
- (D) Urea formaldehyde resin

Q42: Which of the following polymer is used in the manufacture of wood laminates?

- (A) Melamine formaldehyde resin
- (B) cis-poly isoprene
- (C) Phenol and formaldehyde resin
- (D) Urea formaldehyde resin

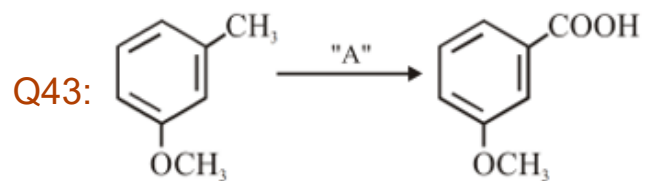
Solution:

Melamine formaldehyde resin is used in the manufacture of wood laminates.



In the above reaction, the reagent "A" is:

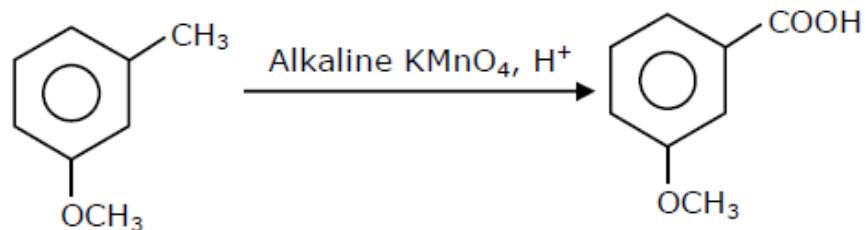
- (A) $\text{NaBH}_4, \text{H}_3\text{O}^+$
- (B) Alkaline $\text{KMnO}_4, \text{H}^+$
- (C) $\text{HCl}, \text{Zn-Hg}$
- (D) LiAlH_4



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- (D) LiAlH_4

Solution:



Q44: Fex_2 and Fey_3 are known when x and y are :

- (A) $x = \text{F, Cl, Br, I}$ and $y = \text{F, Cl, Br, I}$
- (B) $x = \text{F, Cl, Br, I}$ and $y = \text{F, Cl, Br}$
- (C) $x = \text{F, Cl, Br}$ and $y = \text{F, Cl, Br, I}$
- (D) $x = \text{Cl, Br, I}$ and $y = \text{F, Cl, Br, I}$

Q44: $\text{Fe}x_2$ and $\text{Fe}y_3$ are known when x and y are :

- (A) x = F, Cl, Br, I and y = F, Cl, Br, I
- (B) x = F, Cl, Br, I and y = F, Cl, Br
- (C) x = F, Cl, Br and y = F, Cl, Br, I
- (D) x = Cl, Br, I and y = F, Cl, Br, I

Solution:

FeI_3 , does not react because of I^- being very good reducing agent.

Q45: The green house gas/es is (are) :

(A) Carbon dioxide

(B) Oxygen

(C) Water vapour

(D) Methane

Choose the most appropriate answer from the options given below:

(A) (A) and (C) only

(B) (A) and (B) only

(C) (A), (C) and (D) only

(D) (A) only

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(B) (A) and (B) only

☒ (C) (A), (C) and (D) only

(D) (A) only

Solution:

The green-house gases are CO_2 , CH_4 & H_2O vapour.

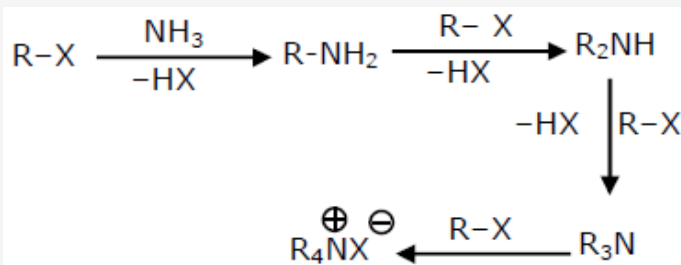
Q46: Ammonolysis of Alkyl halides followed by the treatment with NaOH solution can be used to prepare primary, secondary and tertiary amines. The purpose of NaOH in the reaction is :

- (A) to increase the reactivity of alkyl halide
- (B) to remove basic impurities
- (C) to activate NH_3 used in the reaction
- (D) to remove acidic impurities

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- (C) to activate NH_3 used in the reaction
- (D) to remove acidic impurities

Solution:



During the reaction HX (acid) is formed

Hence, we use NaOH to remove these acidic impurities

Q47: The INCORRECT statements below regarding colloidal solutions is :

- (A) A colloidal solution shows colligative properties.
- (B) An ordinary filter paper can stop the flow of colloidal particles.
- (C) The flocculating power of Al^{3+} is more than that of Na^{+}
- (D) A colloidal solution shows Brownian motion of colloidal particles.

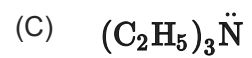
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- (D) A colloidal solution shows Brownian motion of colloidal particles.

Solution:

Colloidal solutions can pass through ordinary filter paper but cannot pass through special filter colloidal solution coated paper.

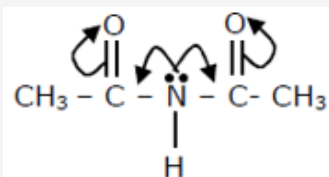
Q48: Which of the following is least basic?



Q48: Which of the following is least basic?

- (A) $(\text{CH}_3\text{CO})_2\ddot{\text{N}}\text{H}$
- (B) $(\text{CH}_3\text{CO})\ddot{\text{N}}\text{HC}_2\text{H}_5$
- (C) $(\text{C}_2\text{H}_5)_3\ddot{\text{N}}$
- (D) $(\text{C}_2\text{H}_5)_2\ddot{\text{N}}\text{H}$

Solution:



Due to high resonance, lp of N is not available for accept H^+
So it is least basic.

Q49: Arrange the following metal complex/compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.

(Atomic numbers Ce = 58, Gd = 64 and Eu = 63.)

(a) $(\text{NH}_4)_2 [\text{Ce}(\text{NO}_3)_6]$

(b) $\text{Gd}(\text{NO}_3)_3$ and

(c) $\text{Eu}(\text{NO}_3)_3$

(A) (b) < (a) < (c)

(B) (a) < (c) < (b)

(C) (c) < (a) < (b)

(D) (a) < (b) < (c)

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(B) (a) < (c) < (b)

(C) (c) < (a) < (b)

(D) (a) < (b) < (c)

Solution:

$(\text{NH}_4)_2 [\text{Ce}(\text{NO}_3)_6]$ ($n = 0$) $\Rightarrow \mu = 0$ B.M

$\text{Eu}(\text{NO}_3)_3$ ($n = 6$) $\Rightarrow \mu = 6.93$ B.M

$\text{Gd}(\text{NO}_3)_3$ ($n = 7$) $\Rightarrow \mu = 7.94$ B.M

Q50: The correct statements about H_2O_2 are :

(A) used in the treatment of effluents.

(B) used as both oxidising and reducing agents.

(C) the two hydroxyl groups lie in the same plane.

(D) miscible with water.

Choose the correct answer from the options given below :

(A) (A), (B) and (D) only

(B) (A), (B), (C) and (D)

(C) (B), (C) and (D) only

(D) (A), (C) and (D) only

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(A) (A), (B) and (D) only

(B) (A), (B), (C) and (D)

(C) (B), (C) and (D) only

(D) (A), (C) and (D) only

Solution:

(1) In H_2O_2 oxidation of oxygen is -1 Therefore acts both as O.A and R.A.

(2) H_2O_2 is miscible in water due to inter molecular H-Bonding.

(3) H_2O_2 has open book structure in which both $-\text{OH}$ group are not in same plane.

(4) It is used in the treatment of effluents.

Q51: A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is ____ min (Round off to the Nearest Integer).

Q51: A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is ____ min (Round off to the Nearest Integer).

108.00

Solution:

1st order
 $A \xrightarrow{\quad} t_{1/2}(A) = 54$

$$A_0 = B_0 = N_0$$

$$A_t = \frac{A_0}{2^{t/54}}$$

1st order
 $B \xrightarrow{\quad} t_{1/2}(B) = 18$

$$B_t = \frac{B_0}{2^{t/18}}$$

$$2^{t/18 - t/54} = 16 = 2^4$$

$$2t/54 = 4$$

$$A_t = 16.B_t$$

$$\frac{A_0}{2^{t/54}} = 16 \times \frac{B_0}{2^{t/18}}$$

$$2^{2t/54} = 16 = 2^4$$

$$t = 108 \text{ min}$$

Q52: $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ absorbs light of wavelength 498 nm during a d–d transition. The octahedral splitting energy for the above complex is _____ $\times 10^{-19}$ J. (Round off to the Nearest Integer). $h = 6.626 \times 10^{-34}$ Js; $c = 3 \times 10^8 \text{ ms}^{-1}$

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4.00

Solution:

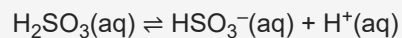
$$\begin{aligned}\Delta_0 &= \frac{hc}{\lambda_{\text{abs}}} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{498 \times 10^{-9}} \\ &= \frac{6.626 \times 3}{498} \times 10^{-17} = 0.0399 \times 10^{-17} = 3.99 \times 10^{-19} \\ &\approx 4 \times 10^{-19} \text{ J}\end{aligned}$$

Q53: Sulphurous acid (H_2SO_3) has $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$. The pH of 0.588 M H_2SO_3 is _____. (Round off to the Nearest Integer).

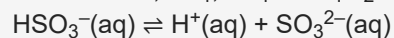
Q53: Sulphurous acid (H_2SO_3) has $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$. The pH of 0.588 M H_2SO_3 is _____. (Round off to the Nearest Integer).

5.00

Solution:



$$0.588 \text{ M} = C, C\alpha_1, C\alpha_1 + C\alpha_1\alpha_2$$



$$C\alpha_1(1 - \alpha_2), C\alpha_1\alpha_2, C\alpha_1\alpha_2$$

+

$$C\alpha_1$$

$$\alpha_1 = \sqrt{\frac{1.7 \times 10^{-2}}{0.588}} = \sqrt{\frac{17}{289 \times 2}}$$

$$\therefore \frac{\alpha_1 < 1}{(1 - \alpha_1) \simeq 1}$$

Hence, $\alpha_2 \ll 1$ and $(1 - \alpha_2) \simeq 1$

$$\therefore [\text{H}^+] = C\alpha_1$$

$$= \sqrt{K_{a1} \times C} = \sqrt{17 \times 10^{-3} \times 0.588}$$

$$= 99.98 \times 10^{-3}$$

$$\text{pH} = 1.99 + 3$$

$$= 4.99 \simeq 5$$

Q54: A 5.0 mol dm^{-3} aqueous solution of KCl has a conductance of 0.55 mS when measured in a cell of cell constant 1.3 cm^{-1} . The molar conductivity of this solution is _____ $\text{mS m}^2 \text{ mol}^{-1}$. (Round off to the Nearest Integer).

Q54: A 5.0 mol dm^{-3} aqueous solution of KCl has a conductance of 0.55 mS when measured in a cell of cell constant 1.3 cm^{-1} . The molar conductivity of this solution is _____ $\text{mSm}^2 \text{ mol}^{-1}$. (Round off to the Nearest Integer).

14.00

Solution:

$$G_{\text{KCl}} = 0.55 \text{ mS} = 55 \times 10^{-5} \text{ S}$$

$$\text{Cell constant} = l/A = 1.3 \text{ cm}^{-1}$$

$$\lambda_{\text{M}} = ?$$

$$R = G(l/A) = 55 \times 10^{-5} \times 1.3 \text{ Scm}^{-1}$$

$$\lambda_{\text{M}} = \frac{K \times 1000}{\text{Molarity}} = \frac{55 \times 1.3 \times 10^{-5} \times 1000}{5 \times 10^{-3}}$$

$$\lambda_{\text{M}} = 11 \times 1.3 \times 10 = 11 \times 13 = 143 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\lambda_{\text{M}} = \frac{143 \times 1000 \times 10^{-3} \text{ S}}{(10^{-2} \text{ M})^{-2}} \text{ mol}^{-1}$$

$$\lambda_{\text{M}} = 143 \times 1000 \times 10^{-4} (\text{m. S}) \text{ m}^2 \cdot \text{mol}^{-1}$$

$$= 14.3$$

$$\text{Ans. } \lambda_{\text{M}} = 14 \text{ Nearest integer}$$

Q55: The number of orbitals with $n = 5$, $m_l = +2$ is _____. (Round off to the Nearest Integer).

Q55: The number of orbitals with $n = 5$, $m_l = +2$ is _____. (Round off to the Nearest Integer).

3.00

Solution:

For $n = 5$

$l = 0, 1, 2, 3, 4$

$l = 2 \rightarrow m = -2, -1, 0, +1, +2$

$l = 3 \rightarrow m = -3, -2, -1, 0, +1, +2, +3$

$l = 4 \rightarrow m = -4, -3, -2, -1, 0, +1, +2, +3, +4$

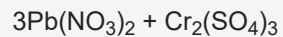
Total no. of orbitals = 3

Q56: When 35 mL of 0.15 M lead nitrate solution is mixed with 20 mL of 0.12 M chromic sulphate solution, _____ $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).

Q56: When 35 mL of 0.15 M lead nitrate solution is mixed with 20 mL of 0.12 M chromic sulphate solution, _____ $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).

525.00

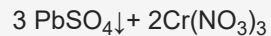
Solution:



35ml, 20 ml

0.15M, 0.12M

= 5.25 m mol, = 2.4 m mol



Moles of PbSO_4 = moles of $\text{Pb}(\text{NO}_3)_2$

= 5.25 m mol

= 525×10^{-5} mol

Answer 525

Q57: In Duma's method of estimation of nitrogen, 0.1840 g of an organic compound gave 30 mL of nitrogen collected at 287 K and 758 mm of Hg pressure. The percentage composition of nitrogen in the compound is _____. (Round off to the Neared Integer).

[Given Aqueous tension at 287 K = 14 mm of Hg]

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[Given Aqueous tension at 287 K = 14 mm of Hg]

19.00

Solution:

$$\text{Moles of N}_2 = \frac{(758-14)}{760} \times \frac{30 \times 10^{-3}}{0.0821 \times 287}$$

$$= 1.246 \times 10^{-3} \text{ mol}$$

$$\text{mass of N} = 1.246 \times 10^{-3} \times 28$$

$$\text{mass \% of 'N'} = \frac{\text{mass of 'N'}}{\text{total mass}} \times 100$$

$$= \frac{1.246 \times 28 \times 10^{-3}}{0.184} \times 100$$

$$= \frac{124.6 \times 28}{0.184} \% = 18.96\%$$

$$\approx 19\%$$

Q58: Ga (atomic mass 70 u) crystallizes in a hexagonal close packed structure. The total number of voids in 0.581 g of Ga is _____ $\times 10^{21}$. (Round off to the Nearest Integer).

[Given: $N_A = 6.023 \times 10^{23}$]

Q58: Ga (atomic mass 70 u) crystallizes in a hexagonal dose packed structure. The total number of voids in 0.581 g of Ga is _____ $\times 10^{21}$. (Round off to the Nearest Integer).
[Given: $N_A = 6.023 \times 10^{23}$]

15.00

Solution:

$$\text{No. of moles of Ga} = \frac{0.581}{70}$$

$$\text{No. of atoms of Ga} = \frac{0.581}{70} \times N_A$$

$$\text{Total number of voids} = \frac{0.581}{70} \times N_A \times 3$$

$$= 0.0249 \times 6 \times 10^{23}$$

$$= 15 \times 10^{21}$$

(As there are one octahedral void and two tetrahedral voids per atom)

Q59: At 363 K, the vapor pressure of A is 21 kPa and that of B is 18 kPa. One mole of A and 2 moles of B are mixed. Assuming that this solution is ideal, the vapor pressure of the mixture is ____ kPa. (Round off to the Nearest Integer).

Q59: At 363 K, the vapor pressure of A is 21 kPa and that of B is 18 kPa. One mole of A and 2 moles of B are mixed. Assuming that this solution is ideal, the vapor pressure of the mixture is ____ kPa. (Round off to the Nearest Integer).

19.00

Solution:

$$\begin{aligned}X_A &= \frac{1}{1+2} = \frac{1}{3} \quad X_B = \frac{2}{3} \\P_A^\circ &= 21 \text{ kPa} \quad P_B^\circ = 18 \text{ kPa} \\P_{\text{total}} &= P_A^\circ X_A + P_B^\circ X_B \\&= 21 \times \frac{1}{3} + 18 \times \frac{2}{3} \\&= 7 + 12 \\&= 19 \text{ kPa}\end{aligned}$$

Q60: At 25°C, 50 g of iron reacts with HCl to form FeCl_2 . The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is (Round off to the Nearest Integer).

[Given: $R=8.314 \text{ J mol}^{-1} \text{ K}^{-1}$. Assume, hydrogen is an ideal gas]

[Atomic mass of Fe is 55.85 u]

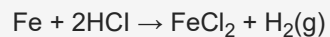
Q60: At 25°C, 50 g of iron reacts with HCl to form FeCl₂. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is (Round off to the Nearest Integer).

[Given: R=8.314 J mol⁻¹ K⁻¹. Assume, hydrogen is an ideal gas]

[Atomic mass of Fe is 55.85 u]

2218.00

Solution:



50g

Moles of Fe = $\frac{50}{55.85}$ mol = moles of H₂

$$W_{\text{irrev}} = -P_{\text{ext}} \cdot \Delta V$$

$$= -\text{moles of H}_2 \times RT$$

$$= -\frac{50}{55.85} \times 8.314 \times 298$$

$$= -2218.05 \text{ J}$$

$$\text{Nearest integer} = 2218$$

Q61: The least value of $|z|$ where z is complex number which satisfies the inequality

$$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7} + 9i|, \quad i = \sqrt{-1}, \text{ is equal to}$$

(A) 8

(B) 3

(C) $\sqrt{5}$

(D) 2

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$$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|} \log_e 2\right) \geq \log_{\sqrt{2}} |5\sqrt{7} + 9i|, \quad i = \sqrt{-1}, \text{ is equal to}$$

(A) 8

(B) 3

(C) $\sqrt{5}$

(D) 2

Solution:

$$2^{\frac{(|z|+3)(|z|-1)}{(|z|+1)}} \geq 2^3$$

$$\Rightarrow \frac{(|z|+3)(|z|-1)}{|z|+1} \geq 3$$

$$\Rightarrow |z|^2 + 2|z| - 3 \geq 3|z| + 3$$

$$\Rightarrow |z|^2 - |z| - 6 \geq 0$$

$$(|z| - 3)(|z| + 2) \geq 0$$

$$|z|_{\min} = 3$$

Q62: Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x which satisfy $\sin^{-1} \left(\frac{3x}{5} \right) + \sin^{-1} \left(\frac{4x}{5} \right) = \sin^{-1} x$ is equal to

- (A) 1
- (B) 3
- (C) 2
- (D) 0

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- (A) 1
- (B) 3
- (C) 2
- (D) 0

Solution:

Taking sine on both side

$$\frac{3x}{5} \sqrt{1 - \frac{16x^2}{25}} + \frac{4x}{5} \sqrt{1 - \frac{9x^2}{25}} = x$$

$$\Rightarrow 3x\sqrt{25 - 16x^2} = 25x - 4x\sqrt{25 - 9x^2}$$

$$\Rightarrow x = 0 \text{ or } 3\sqrt{25 - 16x^2} = 25 - 4\sqrt{25 - 9x^2}$$

$$\Rightarrow 9(25 - 16x^2) = 625 - 200\sqrt{25 - 9x^2} + 16(25 - 9x^2)$$

$$\Rightarrow 200\sqrt{25 - 9x^2} = 800$$

$$\Rightarrow \sqrt{25 - 9x^2} = 4$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

$\therefore$ Total number of solutions = 3

Q63: Let $\alpha \in \mathbb{R}$ be such that the function $f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$ is

continuous at $x = 0$, where $\{x\} = x - [x]$, $[x]$ is the greatest integer less than or equal to x . Then

- (A) $\alpha = 0$
- (B) $\alpha = \frac{\pi}{\sqrt{2}}$
- (C) no such α exists
- (D) $\alpha = \frac{\pi}{4}$

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- (A) $\alpha = 0$
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- (D) $\alpha = \frac{\pi}{4}$

Solution:

$$\lim_{x \rightarrow 0^+} f(x) = f(0) = \lim_{x \rightarrow 0^-} f(x)$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2) \cdot \sin^{-1}(1-x)}{x(1-x)(1+x)}$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2)}{x \cdot 1 \cdot 1} \cdot \frac{\pi}{2}$$

Let $1-x^2 = \cos \theta$

$$\frac{\pi}{2} \lim_{x \rightarrow \theta^+} \frac{\theta}{\sqrt{1-\cos \theta}}$$

$$\frac{\pi}{2} \lim_{x \rightarrow \theta^+} \frac{\theta}{\sqrt{2} \sin \frac{\theta}{2}} = \frac{\pi}{\sqrt{2}}$$

Now, $\lim_{x \rightarrow 0^-} \frac{\cos^{-1}(1-(1+x)^2) \sin^{-1}(-x)}{(1+x)-(1+x)^3}$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2}(-\sin^{-1}x)}{(1+x)(2+x)(-x)}$$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2}}{1 \cdot 2} \cdot \frac{\sin^{-1}x}{x} = \frac{\pi}{4}$$

$\Rightarrow \text{RHL} \neq \text{LHL}$

Function cannot be continuous.

$\Rightarrow$ No value of α exist.

Q64: If the foot of the perpendicular from point (4, 3, 8) on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is (3, 5, 7), then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{6}}$

(D) $\sqrt{\frac{2}{3}}$

Q64: If the foot of the perpendicular from point $(4, 3, 8)$ on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is $(3, 5, 7)$, then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{6}}$

(D) $\sqrt{\frac{2}{3}}$

Solution:

Direction ratio of line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$ is $l, 3, 4$

Let $P(4, 3, 8)$ and $Q(3, 5, 7)$

Now, direction ratio of $PQ \equiv (-1, 2, -1)$

$\therefore PQ \perp L_1$

$$\Rightarrow -l + 6 - 4 = 0$$

$$\Rightarrow l = 2$$

Now point $Q(3, 5, 7)$ satisfies line L_1 ,

$$\Rightarrow a = 1, b = 3$$

Now shortest distance between

$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is

$$d = \frac{(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1})}{|\vec{b_1} \times \vec{b_2}|}$$

$$\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{a_2} - \vec{a_1} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$|\vec{b_1} \times \vec{b_2}| = \sqrt{6}$$

$$d = \frac{(-1+4-2)}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

Q65: Let $A(-1, 1)$, $B(3, 4)$ and $C(2, 0)$ be given three points. A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of $\triangle ABC$ and $\triangle PQC$ respectively, such that $A_1 = 3A_2$, then the value of m is equal to :

(A) 2

(B) 1

(C) $\frac{4}{15}$

(D) 3

Q65: Let $A(-1, 1)$, $B(3, 4)$ and $C(2, 0)$ be given three points. A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of $\triangle ABC$ and $\triangle PQC$ respectively, such that $A_1 = 3A_2$, then the value of m is equal to :

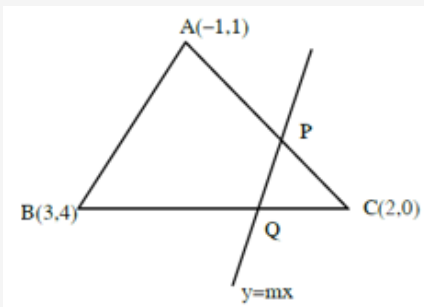
(A) 2

(B) 1

(C) $\frac{4}{15}$

(D) 3

Solution:



$$\text{Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} -1 & 1 & 1 \\ 2 & 0 & 1 \\ 3 & 4 & 1 \end{vmatrix} = \frac{13}{2}$$

$$\text{Equation of line AC is } \rightarrow y - 0 = \frac{-1}{3}(x - 2)$$

$$\text{Equation of line BC is } \rightarrow y - 4 = 4(x - 3)$$

$$P\left(\frac{2}{3m+1}, \frac{2m}{3m+1}\right) \& Q\left(\frac{8}{4-m}, \frac{8m}{4-m}\right)$$

$$\triangle PQC = \frac{1}{2} \begin{vmatrix} 2 & 0 & 1 \\ \frac{2}{3m+1} & \frac{2m}{3m+1} & 1 \\ \frac{8}{4-m} & \frac{8m}{4-m} & 1 \end{vmatrix}$$

$$\frac{13}{6} = \frac{1}{2} \left| \left(\frac{4m}{3m+1} - \frac{16m}{4-m} \right) \right|$$

$$\frac{13}{3} = \left| \frac{52m^2}{11m-3m^2+4} \right| \Rightarrow \frac{1}{3} = \frac{4m^2}{|3m^2-11m-4|}$$

$$\text{Take +ve sign, } 3m^2 - 11m - 4 = 12m^2 \Rightarrow 9m^2 + 11m + 4 = 0 \rightarrow \text{No solution.}$$

$$\text{Take -ve sign, } 3m^2 - 11m - 4 = 12m^2 \Rightarrow 15m^2 - 11m - 4 = 0 \Rightarrow m = 1, \frac{-4}{15}$$

Q66: If $y = y(x)$ is the solution of the differential equation

$\frac{dy}{dx} + (\tan x) y = \sin x$, $0 \leq x \leq \frac{\pi}{3}$, with $y(0) = 0$, then $y\left(\frac{\pi}{4}\right)$ equal to:

- (A) $\frac{1}{2} \log_e 2$
- (B) $\left(\frac{1}{2\sqrt{2}}\right) \log_e 2$
- (C) $\frac{1}{4} \log_e 2$
- (D) $\log_e 2$

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$\frac{dy}{dx} + (\tan x) y = \sin x$, $0 \leq x \leq \frac{\pi}{3}$, with $y(0) = 0$, then $y\left(\frac{\pi}{4}\right)$ equal to:

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(B) $\left(\frac{1}{2\sqrt{2}}\right) \log_e 2$

(C) $\frac{1}{4} \log_e 2$

(D) $\log_e 2$

Solution:

$$\frac{dy}{dx} + y \tan x = \sin x$$

$$I. f. = e^{\int \tan x dx} = e^{\ln \sec x} = \sec x.$$

$$\therefore y \sec x = \int \sin x \cdot \sec x dx$$

$$\Rightarrow y \sec x = \ln |\sec x| + c$$

$$\text{Given } y(0) = 0$$

$$\Rightarrow 0(1) = 0 + c$$

$$\Rightarrow c = 0$$

$$\Rightarrow y \sec x = \ln |\sec x|$$

$$\Rightarrow y = \cos x \ln |\sec x|$$

Now,

$$y\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) \ln\left(\sec \frac{\pi}{4}\right)$$

$$= \frac{1}{\sqrt{2}} \times \ln \sqrt{2}$$

$$= \frac{1}{2\sqrt{2}} \ln 2$$

Q67: Let C be the locus of the mirror image of a point on the parabola $y^2 = 4x$ with respect to the line $y = x$. Then the equation of tangent to C at P(2,1) is :

(A) $x - y = 1$

(B) $2x + y = 5$

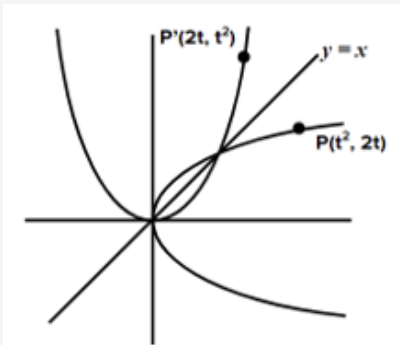
(C) $x + 2y = 4$

(D) $x + 3y = 5$

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- (A) $x - y = 1$
- (B) $2x + y = 5$
- (C) $x + 2y = 4$
- (D) $x + 3y = 5$

Solution:



Let parametric point on $y^2 = 4x$ be $P(t^2, 2t)$

Not image of $P(t^2, 2t)$ about $y = x$ is $P'(2t, t^2)$

So, locus of P' is $x^2 = 4y$ as $x = 2t, y = t^2$

Now tangent of curve $C : x^2 = 4y$ at (2, 1)

$$x \cdot 2 = \frac{4(y+1)}{2}$$

$$x - y - 1 = 0$$

Q68: Let A denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, be divisible by 3. Then probability of event A is equal to

(A) $\frac{9}{56}$

(B) $\frac{11}{27}$

(C) $\frac{3}{7}$

(D) $\frac{4}{9}$

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(A) $\frac{9}{56}$

(B) $\frac{11}{27}$

(C) $\frac{3}{7}$

(D) $\frac{4}{9}$

Solution:

Total cases:

$$\frac{6 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{n(s)} = 6 \cdot 6!$$

Favourable cases:

Number divisible by 3 $\equiv$ Sum of digits must be divisible by 3

Case – I

1, 2, 3, 4, 5, 6

Number of ways = $6 \cdot 5! = 6!$

Case – II

0, 1, 2, 4, 5, 6

Number of ways = $5 \cdot 5!$

Case – III

0, 1, 2, 3, 4, 5

Number of ways = $5 \cdot 5!$

n(favourable) = $6! + 2 \cdot 5 \cdot 5!$

$$P = \frac{6! + 2 \cdot 5 \cdot 5!}{6 \cdot 6!} = \frac{4}{9}$$

Q69: Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, ($a < 0$) be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to :

- (A) $\sqrt{7}$
- (B) $\sqrt{6}$
- (C) $\sqrt{11}$
- (D) $\sqrt{10}$

Q69: Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, ($a < 0$) be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to :

(A) $\sqrt{7}$

(B) $\sqrt{6}$

(C) $\sqrt{11}$

(D) $\sqrt{10}$

Solution:

$$2\sqrt{\frac{a^2}{4} - c} = 2\sqrt{2}, 2\sqrt{a^2 - c} = 2\sqrt{5}$$

$$\frac{a^2}{4} - c = 2$$

$$a^2 = 4c + 8$$

$$3c + 8 = 5$$

$$c = -1, a^2 = 4$$

Equation of tangent

$$y = 2x + c'$$

$$D = \left| \frac{c'}{\sqrt{5}} \right| = r$$

$$\frac{c'}{\sqrt{5}} = \sqrt{\frac{a^2}{4} + a^2 - c}$$

$$\frac{c'}{\sqrt{5}} = \sqrt{1 + 5} = \sqrt{6}$$

$$D = \sqrt{6}$$

Q70: Let $f : S \rightarrow S$ where $S = (0, \infty)$ be a twice differentiable function such that $f(x+1) = xf(x)$. If $g : S \rightarrow \mathbb{R}$ be defined as $g(x) = \log_e f(x)$, then the value of $|g''(5) - g''(1)|$ is equal to :

(A) 1

(B) $\frac{197}{144}$

(C) $\frac{187}{144}$

(D) $\frac{205}{144}$

Q70: Let $f : S \rightarrow S$ where $S = (0, \infty)$ be a twice differentiable function such that $f(x+1) = xf(x)$. If $g : S \rightarrow R$ be defined as $g(x) = \log_e f(x)$, then the value of $|g''(5) - g''(1)|$ is equal to :

(A) 1

(B) $\frac{197}{144}$

(C) $\frac{187}{144}$

(D) $\frac{205}{144}$

Solution:

$$f(x+1) = xf(x)$$

$$\begin{aligned} f(x+N) &= (x+N-1)f(x+N-1) \\ &= (x+N-1)(x+N-2)f(x+N-2) \dots \end{aligned}$$

$$f(x+N) = (x+N-1)(x+N-2) \dots (x-1)(x)f(x)$$

$$g(x+N) = \ln f(x+N) = \ln(x+N-1) + \ln(x+N-2) + \dots + \ln f(x)$$

$$\therefore g'(x+N) = \frac{1}{x+N-1} + \frac{1}{x+N-2} + \dots + \frac{1}{x} + g'(x)$$

$$g''(x+N) - g''(x) = \frac{-1}{(x+N-1)^2} - \frac{1}{(x+N-2)^2} \dots - \frac{1}{x^2}$$

Put $x = 1$ and $N = 4$

$$g''(5) - g''(1) = - \left[\frac{1}{4^2} + \frac{1}{3^2} + \frac{1}{2^2} + \frac{1}{1^2} \right]$$

$$|g''(5) - g''(1)| = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} = \frac{205}{144}$$

Q71: Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from different sides as vertices and β be the number of quadrilaterals having these points from different sides as vertices. Then $(\beta - \alpha)$ is equal to :

- (A) 795
- (B) 1173
- (C) 1890
- (D) 717

Q71: Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from different sides as vertices and β be the number of quadrilaterals having these points from different sides as vertices. Then $(\beta - \alpha)$ is equal to :

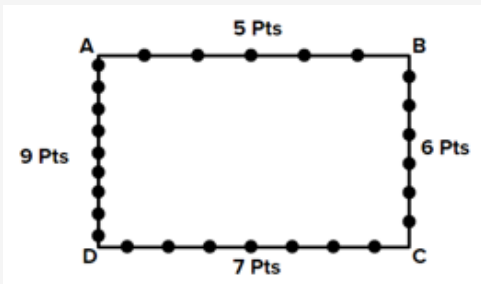
(A) 795

(B) 1173

(C) 1890

(D) 717

Solution:



α = Number of triangles

$$\begin{aligned}\alpha &= 5 \cdot 6 \cdot 7 + 5 \cdot 7 \cdot 9 + 5 \cdot 6 \cdot 9 + 6 \cdot 7 \cdot 9 \\ &= 210 + 315 + 270 + 378 \\ &= 1173\end{aligned}$$

β = Number of quadrilaterals

$$\beta = 5 \cdot 6 \cdot 7 \cdot 9 = 1890$$

$$\beta - \alpha = 1890 - 1173 = 717$$

Q72: Let $P(x) = x^2 + bx + c$ be a quadratic polynomial with real coefficients such that

$\int_0^1 P(x) dx = 1$ and $P(x)$ leaves remainder 5 when it is divided by $(x - 2)$. Then the value of

$9(b + c)$ is equal to

(A) 11

(B) 9

(C) 7

(D) 15

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$\int_0^1 P(x) dx = 1$ and $P(x)$ leaves remainder 5 when it is divided by $(x - 2)$. Then the value of

$9(b + c)$ is equal to

(A) 11

(B) 9

(C) 7

(D) 15

Solution:

$$P(x) = x^2 + bx + c$$

$$P(2) = 5$$

$$5 = 4 + 2b + c \Rightarrow 2b + c = 1 \dots\dots\dots (i)$$

$$\int_0^1 (x^2 + bx + c) dx = 1 \Rightarrow \frac{x^3}{3} + \frac{bx^2}{2} + cx \Big|_0^1 = 1$$

$$\frac{b}{2} + c = \frac{2}{3} \Rightarrow b + 2c = \frac{4}{3} \dots\dots\dots (ii)$$

After solving (i) & (ii) we get

$$b = \frac{2}{9} \& c = \frac{5}{9}$$

$$9(b + c) = 9\left(\frac{2}{9} + \frac{5}{9}\right) = 7$$

Q73: Let C_1 be the curve obtained by the solution of differential equation

$2xy \frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the solution of $\frac{2xy}{x^2 - y^2} = \frac{dy}{dx}$. If both the curves pass through $(1,1)$, then the area enclosed by the curves C_1 and C_2 is equal to

- (A) $\frac{\pi}{2} - 1$
- (B) $\pi + 1$
- (C) $\frac{\pi}{4} + 1$
- (D) $\pi - 1$

Q73: Let C_1 be the curve obtained by the solution of differential equation

$2xy \frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the solution of $\frac{2xy}{x^2 - y^2} = \frac{dy}{dx}$. If both the curves pass through $(1,1)$, then the area enclosed by the curves C_1 and C_2 is equal to

(A) $\frac{\pi}{2} - 1$

(B) $\pi + 1$

(C) $\frac{\pi}{4} + 1$

(D) $\pi - 1$

Solution:

$$C_1 : \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v} \Rightarrow x \frac{dv}{dx} = \frac{-(1+v^2)}{2v}$$

$$\int \left(\frac{2v}{1+v^2} \right) dv = - \int \frac{dx}{x} \Rightarrow \ln(1+v^2)x = c$$

$$C_1 : \frac{(x^2+y^2)}{x} = c$$

Since the curve passing through (1,1), gives $c = 2 \Rightarrow C_1 : x^2 + y^2 = 2x$

$$C_2 : \frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{2v}{1-v^2} \Rightarrow x \frac{dv}{dx} = \frac{v+v^3}{1-v^2}$$

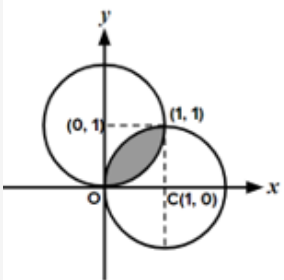
$$\frac{1-v^2}{v(1+v^2)} dv = \frac{dx}{x} \Rightarrow \int \left(\frac{1}{v} - \frac{2v}{1+v^2} \right) dv = \int \frac{dx}{x}$$

$$\ln \left(\frac{v}{1+v^2} \right) = \ln x + c \Rightarrow \frac{y}{x^2+y^2} = c$$

Since the curve passing through (1,1), gives $c = \frac{1}{2} \Rightarrow \frac{y}{x^2+y^2} = \frac{1}{2}$

$$C_2 : x^2 + y^2 = 2y$$

$\therefore$ Area bounded between $(x-1)^2 + y^2 = 1$ and $x^2 + (y-1)^2 = 1$ is



$$\text{Area} = \int_0^1 (\sqrt{2x-x^2} - \sqrt{1-x^2} - 1) dx$$

$$\begin{aligned}
&= \left[\frac{(x-1)}{2} \sqrt{2x-x^2} - \frac{1}{2} \sin^{-1}(x-1) - \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x - x \right]_0^1 \\
&= \left(\frac{\pi}{4} - 1 \right) - \left(\frac{-\pi}{4} \right) = \frac{\pi}{2} - 1
\end{aligned}$$

Q74: Let $A = \{2, 3, 4, 5, \dots, 30\}$ and ' $\simeq$ ' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to :

- (A) 6
- (B) 5
- (C) 8
- (D) 7

Q74: Let $A = \{2, 3, 4, 5, \dots, 30\}$ and ' $\simeq$ ' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to :

(A) 6

(B) 5

(C) 8

(D) 7

Solution:

$$A = \{2, 3, 4, \dots, 30\}; \frac{a}{b} = \frac{c}{d}$$

$$\therefore (4, 3) = \frac{4}{3} = \frac{8}{6} = \frac{12}{9} = \frac{16}{12} = \frac{20}{15} = \frac{24}{18} = \frac{28}{21}$$

$\therefore$ Total number of elements = 7

Q75: If the points of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b$, $b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to

- (A) 10
- (B) 5
- (C) 12
- (D) 6

Q75: If the points of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b$, $b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to

(A) 10

(B) 5

(C) 12

(D) 6

Solution:

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

$$x^2 + y^2 = 4b$$

$$y^2 = 3x^2$$

$$\Rightarrow b = x^2$$

$$y^2 = 3b$$

$$\therefore \frac{b}{16} + \frac{3}{b} = 1$$

$$\Rightarrow b^2 - 16b + 48 = 0$$

$$\Rightarrow b = 4, 12$$

$$\therefore b > 4 \Rightarrow b = 12$$

Q76: If (x, y, z) be an arbitrary point lying on a plane P which passes through the points $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of the expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$

Is equal to

- (A) 3
- (B) 0
- (C) -45
- (D) 39

Q76: If (x, y, z) be an arbitrary point lying on a plane P which passes through the points $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of the expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$

Is equal to

(A) 3

(B) 0

(C) -45

(D) 39

Solution:

Equation of plane is $x + y + z = 42$

$$\Rightarrow (x - 11) + (y - 19) + (z - 12) = 0$$

Let $x - 11 = u$; $y - 19 = v$; $z - 12 = w$

i.e. $u + v + w = 0$

$$\therefore \frac{u}{v^2 \cdot w^2} + \frac{v}{u^2 \cdot w^2} + \frac{w}{u^2 \cdot v^2} - \frac{3}{u \cdot v \cdot w} + 3$$

$$= \left(\frac{u^3 + v^3 + w^3 - 3uvw}{u^2 v^2 w^2} \right) + 3 = 3$$

$$[\because u + v + w = 0 \Rightarrow u^3 + v^3 + w^3 = 3uvw]$$

Q77: The maximum value of $f(x) = \begin{vmatrix} \sin^2 x & 1 + \cos^2 x & \cos 2x \\ 1 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}, x \in R$ is

(A) $\sqrt{5}$

(B) 5

(C) $\frac{3}{4}$

(D) $\sqrt{7}$

Q77: The maximum value of $f(x) = \begin{vmatrix} \sin^2 x & 1 + \cos^2 x & \cos 2x \\ 1 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}, x \in R$ is

(A) $\sqrt{5}$

(B) 5

(C) $\frac{3}{4}$

(D) $\sqrt{7}$

Solution:

$$C_1 \rightarrow C_1 + C_2$$

$$\begin{vmatrix} 2 & 1 + \cos^2 x & \cos 2x \\ 2 & \cos^2 x & \cos 2x \\ 1 & \cos^2 x & \sin 2x \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{vmatrix} 0 & 1 & 0 \\ 2 & \cos^2 x & \cos 2x \\ 1 & \cos^2 x & \sin 2x \end{vmatrix}$$

Open w.r.t. R_1

$$-(2 \sin 2x - \cos 2x)$$

$$\cos 2x - 2 \sin 2x = f(x)$$

$$f(x)|_{\max} = \sqrt{1 + 4} = \sqrt{5}$$

Q78: Let

$\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$, $\alpha \in \mathbb{R}$, then the value of $\alpha + |\vec{r}|^2$ is equal to

- (A) 13
- (B) 15
- (C) 9
- (D) 11

Q78: Let

$\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$, $\alpha \in \mathbb{R}$, then the value of $\alpha + |\vec{r}|^2$ is equal to

(A) 13

(B) 15

(C) 9

(D) 11

Solution:

$$\vec{r} \times \vec{a} - \vec{b} \times \vec{r} = \vec{0}$$

$$\Rightarrow \vec{r} \times \vec{a} + \vec{r} \times \vec{b} = \vec{0}$$

$$\Rightarrow \vec{r} \times (\vec{a} + \vec{b}) = \vec{0}$$

$$\Rightarrow \vec{r} = \lambda (\vec{a} + \vec{b}) = \lambda (3\hat{i} - \hat{j} + 2\hat{k})$$

$$\text{Also } \vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3,$$

$$\therefore \lambda (3\alpha - 2 + 2) = 3$$

$$\therefore \lambda\alpha = 1$$

$$\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$$

$$\lambda (6 - 5 - 2\alpha) = -1$$

$$\lambda (1 - 2\alpha) = -1$$

$$\lambda - 2 = -1$$

$$\lambda = 1$$

$$\lambda\alpha = 1 \Rightarrow \alpha = 1$$

$$\vec{r} = 3\hat{i} - \hat{j} + 2\hat{k}$$

$$\therefore a + |\vec{r}|^2 = 1 + (9 + 1 + 4) = 15$$

Q79: Let f be a real valued function, defined on $\mathbb{R} - \{-1, 1\}$ and given by

$$f(x) = 3\log_e \left| \frac{x-1}{x+1} \right| - \frac{2}{x-1}$$

Then in which of the following intervals, function $f(x)$ is increasing ?

- (A) $(-\infty, \infty) - \{-1, 1\}$
- (B) $(-\infty, -1) \cup ([\frac{1}{2}, \infty) - \{1\})$
- (C) $(-1, \frac{1}{2}]$
- (D) $(-\infty, \frac{1}{2}] - \{-1\}$

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Solution:

$$\begin{aligned} f'(x) &= 3 \left(\frac{x+1}{x-1} \right) \frac{2}{(x+1)^2} + \frac{2}{(x-1)^2} \\ &= \frac{6}{x^2-1} + \frac{2}{(x-1)^2} \\ &= \frac{6(x-1)+2(x+1)}{(x-1)^2(x+1)} \\ f'(x) &= \frac{8x-4}{(x+1)(x-1)^2} = \frac{4(2x-1)}{(x+1)(x-1)^2} \\ \therefore f'(x) &\geq 0 \forall x \in (-\infty, -1) \cup [\frac{1}{2}, 1) \cup (1, \infty) \end{aligned}$$

Q80: Consider the integral

$$I = \int_0^{10} \frac{[x]e^{[x]}}{e^x - 1} dx$$

where $[x]$ denotes the greatest integer less than or equal to x . Then the value of I is equal to :

- (A) $9(e - 1)$
- (B) $9(e + 1)$
- (C) $45(e + 1)$
- (D) $45(e - 1)$

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- (A) $9(e - 1)$
- (B) $9(e + 1)$
- (C) $45(e + 1)$
- (D) $45(e - 1)$

Solution:

$$\begin{aligned} I &= \int_0^{10} \frac{[x]e^{[x]}}{e^x - 1} dx \\ &= e \left[e \int_1^2 e^{-x} dx + 2e^2 \int_2^3 e^{-x} dx + \dots + 9e^9 \int_9^{10} e^{-x} dx \right] \\ &= -e \left[e(e^{-2} - e^{-1}) + 2e^2(e^{-3} - e^{-2}) + 3e^3(e^{-4} - e^{-3}) + \dots + 9e^9(e^{-10} - e^{-9}) \right] \\ &= -e \left[(e^{-1} - 1) + 2(e^{-1} - 1) + 3(e^{-1} - 1) + \dots + 9(e^{-1} - 1) \right] \\ &= -e [45e^{-1} - 45] = 45[e - 1] \end{aligned}$$

Q81: In $\triangle ABC$, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of $\triangle ABC$ is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of $\triangle ABC$, then the value of $2R + r$ (in cm) is equal to _____.

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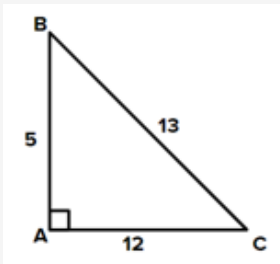
15.00

Solution:

Let $c = 5, b = 12, \Delta = 30$

$$\Delta = \frac{1}{2} \cdot c \cdot b \cdot \sin A = \frac{1}{2} \times 5 \times 12 \cdot \sin A = 30$$

$$\therefore \sin A = 1 \Rightarrow A = 90 \Rightarrow a = 13$$



$$\Rightarrow 2R = \frac{a}{\sin A} = 13; r = \frac{\Delta}{s} = \frac{30}{15} = 2$$

$$\therefore 2R + r = 13 + 2 = 15$$

Q82: Let

$S_n(x) = \log_{a^{\frac{1}{2}}} x + \log_{a^{\frac{1}{3}}} x + \log_{a^{\frac{1}{6}}} x + \log_{a^{\frac{1}{11}}} x + \log_{a^{\frac{1}{18}}} x + \log_{a^{\frac{1}{27}}} x + \dots$ upto
n-terms where $a > 1$. If $S_{24}(x) = 1093$ and $S_{12}(2x) = 265$, then the value of a is equal to

_____.

Q82: Let

$S_n(x) = \log_{a^{\frac{1}{2}}} x + \log_{a^{\frac{1}{3}}} x + \log_{a^{\frac{1}{6}}} x + \log_{a^{\frac{1}{11}}} x + \log_{a^{\frac{1}{18}}} x + \log_{a^{\frac{1}{27}}} x + \dots$ upto
n-terms where $a > 1$. If $S_{24}(x) = 1093$ and $S_{12}(2x) = 265$, then the value of a is equal to

_____.

16.00

Solution:

$$S_n(x) = 2\log_a x + 3\log_a x + 6\log_a x + 11\log_a x + \dots$$

$$S_n(x) = \log_a x (2 + 3 + 6 + 11 + \dots)$$

$$\text{Let } S_r = 2 + 3 + 6 + 11 + \dots$$

$$\text{General term } T_r = r^2 - 2r + 3$$

$$S_n(x) = \sum_{r=1}^n \log_a x (r^2 - 2r + 3)$$

$$S_{24}(x) = \sum_{r=1}^{24} \log_a x (r^2 - 2r + 3)$$

$$S_{24}(x) = \log_a x \times \sum_{r=1}^{24} (r^2 - 2r + 3)$$

$$1093 = 4972\log_a x$$

$$\log_a x = \frac{1}{4}$$

$$x = a^{\frac{1}{4}}$$

$$S_{12}(2x) = \log_a(2x) \sum_{r=1}^{12} (r^2 - 2r + 3)$$

$$265 = 530\log_a(2x)$$

$$\log_a(2x) = \frac{1}{2}$$

$$2x = a^{\frac{1}{2}} \dots \dots \dots (ii)$$

After solving (i) and (ii), we get

$$a^{\frac{1}{4}} = 2$$

$$a = 16$$

Q83: For the real numbers α, β, γ and δ if

$$\int \frac{(x^2-1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)} dx == \alpha \log_e \left(\tan^{-1} \left(\frac{x^2+1}{x} \right) \right) + \beta \tan^{-1} \left(\frac{\gamma(x^2-1)}{x} \right) + \delta \tan^{-1} \left(\frac{x^2+1}{x} \right)$$

where C is an arbitrary constant, then the value of $10(\alpha + \beta\gamma + \delta)$ is equal to ____

Q83: For the real numbers α, β, γ and δ if

$$\int \frac{(x^2-1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)} dx == \alpha \log_e \left(\tan^{-1} \left(\frac{x^2+1}{x} \right) \right) + \beta \tan^{-1} \left(\frac{\gamma(x^2-1)}{x} \right) + \delta \tan^{-1} \left(\frac{x^2+1}{x} \right) + C$$

where C is an arbitrary constant, then the value of $10(\alpha + \beta\gamma + \delta)$ is equal to ____

Solution:

$$\begin{aligned}
 & \int \frac{(x^2-1) + \tan^{-1}\left(\frac{x^2+1}{x}\right) dx}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right) dx} \\
 &= \int \frac{(x^2-1) dx}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)} + \int \frac{\tan^{-1}\left(x+\frac{1}{x}\right) dx}{(x^4+3x^2+1)\tan^{-1}\left(x+\frac{1}{x}\right)} \\
 &= \int \frac{\left(1-\frac{1}{x^2}\right) dx}{\left(\left(x+\frac{1}{x}\right)^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)} + \int \frac{dx}{(x^4+3x^2+1)} \\
 &= I_1 + I_2
 \end{aligned}$$

For I_1

Put

$$x + \frac{1}{x} = t, \left(1 - \frac{1}{x^2}\right) dx = dt$$

$$I_1 = \int \frac{dt}{(t^2+1)\tan^{-1}(t)}$$

Gain $\tan^{-1} = k$

$$\left(\frac{1}{t^2+1}\right) dt = dk$$

$$I_1 = \int \frac{dk}{k} = \ln k + c = \ln(\tan^{-1} t) + c$$

$$= \ln\left(\tan^{-1} x + \frac{1}{x}\right) + c$$

$$= \ln\left(\tan^{-1}\left(\frac{x^2+1}{x}\right)\right) + c$$

$$I_2 = \int \frac{dx}{x^4+3x^2+1}$$

$$= \int \frac{dx}{x^2\left(x^2+\frac{1}{x^2}+3\right)}$$

$$= \int \frac{\frac{1}{x^2} dx}{\left(x^2+\frac{1}{x^2}\right)+3}$$

$$= \frac{1}{2} \int \frac{\left(\frac{1}{x^2}-1\right) dx}{\left(x+\frac{1}{x}\right)^2+1} + \frac{1}{2} \int \frac{\left(1+\frac{1}{x^2}\right) dx}{\left(x-\frac{1}{x}\right)^2+5}$$

$$\begin{aligned}
&= -\frac{1}{2}\tan^{-1}\left(x + \frac{1}{x}\right) + \frac{1}{2}\tan^{-1}\left(\frac{x - \frac{1}{x}}{\sqrt{5}}\right) \\
&= \frac{-1}{2}\tan^{-1}\left(\frac{x^2+1}{x}\right) + \frac{1}{2\sqrt{5}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{5}x}\right) \\
\therefore I &= \ln\left(\tan^{-1}\left(\frac{x^2+1}{x}\right)\right) - \frac{1}{2}\tan^{-1}\left(\frac{x^2+1}{x}\right) + \frac{1}{2\sqrt{5}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{5}x}\right) + c \\
\therefore \alpha &= 1, \beta = \frac{1}{2\sqrt{5}}, \gamma = \frac{1}{\sqrt{5}}, \delta = \frac{-1}{2} \\
10(\alpha + \beta\gamma + \delta) &= 10\left(1 + \frac{1}{10} - \frac{1}{2}\right) = 6
\end{aligned}$$

Q84: If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to ____.

Q84: If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to ____.

2.00

Solution:

$$DC \text{ of line} \equiv \left(\frac{3}{\sqrt{m^2+10}}, \frac{-m}{\sqrt{m^2+10}}, \frac{1}{\sqrt{m^2+10}} \right)$$

$$Q \equiv \left(1 + \frac{3r}{\sqrt{m^2+10}}, -2 + \frac{-mr}{\sqrt{m^2+10}}, 3 + \frac{r}{\sqrt{m^2+10}} \right)$$

$$Q \text{ lies on } x + 2y - 3z + 10 = 0$$

$$1 + \frac{3r}{\sqrt{m^2+10}} - 4 - \frac{2mr}{\sqrt{m^2+10}} - 9 - \frac{3r}{\sqrt{m^2+10}} + 10 = 0$$

$$\Rightarrow \frac{r}{\sqrt{m^2+10}} (3 - 2m - 3) = 2$$

$$\Rightarrow r^2 m^2 = m^2 + 10$$

$$\Rightarrow \frac{7}{2} m^2 = m^2 + 10$$

$$\Rightarrow m^2 = 4$$

$$\Rightarrow |m| = 2$$

Q85: Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real numbers such that $A = XB$, where

$X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and $k \in \mathbb{R}$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$, then the value of k is _____.

Q85: Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real numbers such that $A = XB$, where

$X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and $k \in R$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$, then the value of k is _____.

1.00

Solution:

$$A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$$

$$A = XB$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} b_1 - b_2 \\ b_1 + kb_2 \end{bmatrix}$$

$$\therefore a_1 = \frac{b_1 - b_2}{\sqrt{3}}; a_2 = \frac{b_1 + kb_2}{\sqrt{3}}$$

$$a_1^2 + a_2^2 = \frac{b_1^2 + b_2^2 - 2b_1b_2 + b_1^2 + k^2b_2^2 + 2kb_1b_2}{3} = 2 \frac{(b_1^2 + b_2^2)}{3}$$

$$b_2^2 = kb_2^2 + 2b_1b_2(k - 1)$$

$$\Rightarrow k = 1$$

Q86: Let n be a positive integer. Let

$$A = \sum_{k=0}^n (-1)^k {}^n C_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$$

If $63A = 1 - \frac{1}{2^{30}}$, then n is equal to ____.

Q86: Let n be a positive integer. Let

$$A = \sum_{k=0}^n (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$$

If $63A = 1 - \frac{1}{2^{30}}$, then n is equal to ____.

6.00

Solution:

$$\begin{aligned} A &= \sum_{k=0}^n (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \dots + \left(\frac{31}{32}\right)^k \right] \\ &= \left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n + \left(\frac{1}{8}\right)^n + \left(\frac{1}{16}\right)^n + \left(\frac{1}{32}\right)^n \\ A &= \frac{\left(\frac{1}{2}\right)^n \left[1 - \left(\frac{1}{2}\right)^{5n} \right]}{1 - \left(\frac{1}{2}\right)^n} \end{aligned}$$

$$\text{When } n = 6 \Rightarrow 63A = 1 - \frac{1}{2^{30}} \Rightarrow n = 6$$

Q87: Let $\vec{c}$ be a vector perpendicular to the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8$ then the value of $\vec{c} \cdot (\vec{a} \times \vec{b})$ is equal to ____

Q87: Let $\vec{c}$ be a vector perpendicular to the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8$ then the value of $\vec{c} \cdot (\vec{a} \times \vec{b})$ is equal to ____

28.00

Solution:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 3\hat{i} - 2\hat{j} + \hat{k}$$

Since $\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b}$.

$$\text{Let } \vec{c} = \lambda (3\hat{i} - 2\hat{j} + \hat{k})$$

$$\Rightarrow \vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8 \Rightarrow \lambda (3 - 2 + 3) = 8 \Rightarrow \lambda = 2$$

$$\vec{c} \cdot (\vec{a} \times \vec{b}) = \lambda |\vec{a} \times \vec{b}|^2 = 2(9 + 4 + 1) = 28$$

Q88: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x + a, & x < 0 \\ |x - 1|, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x + 1, & x < 0 \\ (x - 1)^2 + b, & x \geq 0 \end{cases}$$

where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in \mathbb{R}$, then $a + b$ is equal to ____.

Q88: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

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where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in \mathbb{R}$, then $a + b$ is equal to ____.

1.00

Solution:

$$\begin{aligned} g \circ f(x) &= \begin{cases} f(x) + 1 & \text{if } f(x) < 0 \\ \{f(x) - 1\}^2 + b & \text{if } f(x) \geq 0 \end{cases} \\ &= \begin{cases} x + a + 1 & \text{if } x < -a \\ (x + a - 1)^2 + b & \text{if } -a \leq x < 0 \\ (|x - 1| - 1)^2 + b & \text{if } x \geq 0 \end{cases} \end{aligned}$$

Also $g \circ f(x)$ is continuous at $x = -a$

$$g \circ f(0) = g \circ f(0^+) = g \circ f(0^-)$$

$$\Rightarrow b = b = (a - 1)^2 + b$$

$$\Rightarrow a = 1$$

$$\text{Hence, } a + b = 1 + 0 = 1$$

Q89: Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to ____.

Q89: Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to ____.

5.00

Solution:

$$\text{For group - 1 : } \frac{\sum x_i}{10} = 2 \Rightarrow \sum x_i = 20$$

$$\frac{\sum x_i^2}{10} - (2)^2 = 2 \Rightarrow \sum x_i^2 = 60$$

$$\text{For group - 2 : } \frac{\sum y_i}{n} = 3 \Rightarrow \sum y_i = 3n$$

$$\frac{\sum y_i^2}{n} - 3^2 = 1 \Rightarrow \sum y_i^2 = 10n$$

Now, combined variance

$$\sigma^2 = \frac{\sum (x_i^2 + y_i^2)}{10+n} - \left(\frac{\sum (x_i + y_i)}{10+n} \right)^2$$

$$\Rightarrow \frac{17}{9} = \frac{60+10n}{10+n} - \frac{(20+3n)^2}{(10+n)^2}$$

$$\Rightarrow 17(n^2 + 20n + 100) = 9(n^2 + 40n + 200)$$

$$\Rightarrow 8n^2 - 20n - 100 = 0$$

$$\Rightarrow 2n^2 - 5n - 25 = 0$$

$$\Rightarrow n = 5$$

Q90: Let $\frac{1}{16}$, a and b in G.P. $\frac{1}{a}$, $\frac{1}{b}$, 6 are in A.P., where $a, b > 0$. Then $72(a + b)$ is equal to

Q90: Let $\frac{1}{16}, a$ and b in G.P. $\frac{1}{a}, \frac{1}{b}, 6$ are in A.P., where $a, b > 0$. Then $72(a + b)$ is equal to

14.00

Solution:

$$a^2 = \frac{b}{16}; \frac{2}{b} = \frac{1}{a} + 6 \Rightarrow b = \frac{2a}{1+6a}$$

$$\therefore 16a^2 = \frac{2a}{1+6a} \Rightarrow 96a^2 + 16a - 2 = 0$$

$$48a^2 + 8a - 1 = 0 \Rightarrow 48a^2 + 12a - 4a - 1 = 0$$

$$12a(4a + 1) - (4a + 1) = 0 \Rightarrow a = \frac{-1}{4}, \frac{1}{12} \Rightarrow b = 1, \frac{1}{9}$$

$$\therefore 72(a + b) = 54 \text{ or } 14$$

Since a cannot be negative, then $72(a + b) \neq 54$. Hence correct answer is 14.